

Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

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Abstract

We repair an option-only mean–variance implementation and reassess what its historical evidence can support. Option positions are premium-funded, expiring cashflows. For the fixed Greek loading matrix, the covariance retains factor/residual cross terms rather than assuming an OLS projection that the code does not estimate. The repaired R1 allocation maximizes expected return net of asymmetric costs and a variance penalty, permits cash, and imposes CVaR, stress, margin, collateral, assignment, integer, and point-in-time liquidity limits before a position is accepted. The covariance factors are limited to delta, centered quadratic spot returns, and vega; skew and tail enter premia and stress tests. We make no claim that these standard mathematical ingredients are new. The original E1 Sharpes are retained as retrospective development evidence, not headline proof: E1 was selected after inspection of the post-2020 window, and its VIX-enabled CPCV wealth paths are absorbed at zero in March 2020 despite positive return-based Sharpe. Under R1’s hard survival gate that is a failure. All existing observations are labeled development data, and confirmatory claims require 36 future untouched monthly returns. The separately versioned R1.1 development arm raises the risk ceiling to 25 percent, pursues but does not force 50 percent gross, and specifies a close-to-next-open VIX-40 exit rule. Its event-date return remains unscored until matched licensed CBBO is available; its EGARCH overlay fails a prespecified promotion gate.

Key findings and claim boundary

- R1 includes the covariance cross terms omitted by E1 and prices predictable costs inside a uniquely scaled utility objective.
- Gross premium is an upper bound; cash replaces split-leg “burn” and infeasible integer books fail closed to cash.
- Any zero-wealth, margin, collateral, or integer-feasibility failure overrides Sharpe. The legacy VIX CPCV paths therefore fail the repaired survival standard.
- Existing results are retrospective development evidence. A prospective freeze manifest requires 36 untouched monthly observations for confirmation.
- R1.1 does not attain its 50 percent deployment target under the retained constraints; its VIX risk-off arm is unscored without executable event quotes.
- The licensed-quote audit leaves headline R1.1 orig+VIX Sortino at 2.275 under modeled costs and 2.052 under observed touch fills, with 94.1 percent entry coverage.
- R1.1 is the paper’s final development model; R2 remains a gate-failed diagnostic.

1. Research question and scope

The paper asks whether an option-only allocator can combine premium accounting, Greeks, expiry, margin, and displayed liquidity into a portfolio procedure that survives realistic frictions. The evidence available for answering that question is entirely retrospective. It is therefore development evidence, not a confirmatory holdout or a live track record.

The repaired specification, R1, makes three changes to the earlier E1 development model. First, it retains the generally nonzero covariance between Greek factors and Greek residuals. Second, it replaces maximum Sharpe with cost-aware mean–variance utility and allows the optimizer to hold cash. Third, survival, collateral, margin, and integer feasibility are hard gates: an absorbed-zero path cannot be rescued by a positive arithmetic Sharpe ratio. R1 uses only delta, centered squared-return, and implied-volatility factors in covariance. Skew and tail information enters expected returns and stress tests, not an exposure matrix that the code does not estimate.

The mathematical ingredients are standard or direct adaptations: Taylor Greek maps, covariance propagation, mean–variance utility, CVaR, and convex constraints. We do not claim a new portfolio theorem. The potential contribution is operational and empirical: putting those ingredients into one auditable, point-in-time option workflow and reporting where that workflow fails. The earlier E1 results are retained below as legacy development evidence because they document the research path; they are not the repaired headline model.

Capacity remains a binding finding in the local data. The tested contracts do not support mechanical scaling from roughly \$1 million to \$5 million NAV. Multi-exchange routing could change that conclusion, but the present data contain neither routed orders nor fills, so this paper does not claim that routing solves the constraint.

1.1. Relation to the literature

We adapt mean–variance portfolio choice (Markowitz, 1952; Sharpe, 1966) to premium-funded option cashflows. Prior option-portfolio work already studies optimized option allocations, Greek-based risk maps, transaction costs, and mean–variance structure (Faias and Santa-Clara, 2017; Zhao and Palomar, 2018; Malamud, 2014; Liang et al., 2008). Gross-exposure regularization and robust conic portfolio choice also have established foundations (Fan et al., 2012; Goldfarb and Iyengar, 2003). The breadth and estimation-error mechanics follow the familiar tradeoff between choice-set size and input quality (Grinold and Kahn, 2000; DeMiguel et al., 2009). The expected-return channels come from the empirical option-return literature: expected option returns and their signs (Coval and Shumway, 2001), delta-hedged gains and the volatility risk premium (Bakshi and Kapadia, 2003), variance risk premia (Carr and Wu, 2009; Bollerslev et al., 2009), index option portfolio evidence (Driessen and Maenhout, 2007; Broadie et al., 2009), direct option-portfolio optimization (Faias and Santa-Clara, 2017), and the embedded-leverage demand that makes cheap convexity systematically expensive (Frazzini and Pedersen, 2022). The validation protocol draws on the backtest-overfitting and multiple-testing literature (White, 2000; Hansen, 2005; Bailey et al., 2017; Bailey and López de Prado, 2014; López de Prado, 2018). The warning that option-like pay-offs can manufacture attractive Sharpe ratios while retaining rare crash losses is central here (Goetzmann et al., 2002).

Relative to this literature, the marginal contribution sought here is the integrated implementation: contract-level premium accounting, a complete Greek/residual joint covariance, pre-trade costs and liquidity caps, explicit collateral and margin, exact versus proxied spread accounting, and VIX settlement handling. None is novel in isolation. The paper supplies self-contained derivations to document the implementation, not to claim priority for standard mathematics.

1.2. Outline

Section 2 develops the theory: premium accounting, the Greek-induced covariance decomposition, the structural expected-return channels with the breadth results that motivate them, and the constrained conic allocation. Section 3 describes the four implementation universes, repaired R1 procedure, legacy E1 development experiment, transaction-cost stack, and validation protocol. Section 4 reports gross and net results against both baselines and the mechanism evidence. Section 5 presents the distributional and multiple-testing evidence, including the cross-validation stress analysis. Section 6 states the capacity and execution limits of the claim, and Appendix A collects proofs and solver details. Every figure and table is generated from machine-readable artifacts, and an independent verification harness re-audits the reported numbers, settlement handling, and spread-source policy on each build.

2. Theory: option allocation as funded cashflows

The object of the paper is an option book, not a transformed equity book. An option contract i at decision date t has a mark $C_{i,t}$, multiplier m_i , payoff rule, settlement convention, Greeks, displayed liquidity, and a forecast of conditional option compensation. A portfolio weight w_i is the premium notional committed to that contract divided by NAV. These are *premium weights*. A long option with $w_i = 0.03$ can lose at most three percent of NAV before trans-

action costs and financing; a short option with $w_i = -0.03$ receives three percent of NAV in premium but carries a future mark-to-market and settlement liability. Cash is the collateral numeraire.

This distinction is the main mathematical reason the option-only problem differs from ordinary stock Markowitz. Stock weights usually satisfy a unit-budget equation. Option premium weights do not. Residual cash, margin collateral, and unpaid short-option losses are separate accounting objects. The correct bridge from theory to implementation is therefore to write option cashflows explicitly, build a Greek-induced covariance for those cashflows, and then constrain the resulting allocation by premium, stress, margin, and liquidity budgets.

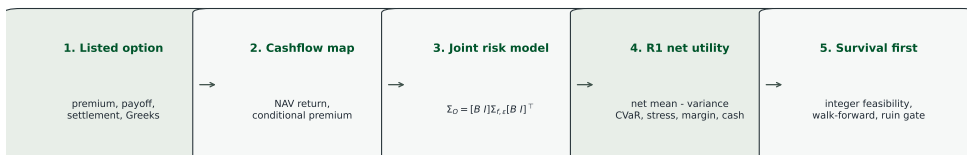


Figure 1: Research pipeline. The theory starts from listed option contracts, turns them into NAV-normalized cashflows, estimates the complete Greek/residual joint covariance, prices predictable costs inside R1 utility, and subjects the integer book to operational and survival tests.

2.1. Why ordinary stock Markowitz is insufficient

Ordinary Markowitz (Markowitz, 1952) starts from returns on assets that can be held with weights summing to one. The budget equation does much of the economic work: it fixes scale, makes positive and negative weights interpretable as capital allocation and borrowing, and leaves the covariance matrix as the main risk object. Listed options break that simplification. Premium paid is not the same as capital at risk for a short option; margin required is not the same as premium received; and a low-premium option can carry large delta, gamma, vega, and gap exposure. A variance penalty alone cannot prevent the optimizer from selecting positions that are cheap by premium but expensive by stress loss or market depth.

The right mathematical analogue is therefore not a simple replacement of stock returns by option returns. It is a funded cashflow problem. Premium weights determine the mark-to-market contribution. Collateral and margin constraints determine whether the position can be carried. Greek and stress constraints determine whether local variance misses a bad state. Liquidity caps determine whether the target can be traded at the account size. The portfolio problem becomes a constrained allocation over option cashflow claims, with cash and collateral as the reference asset.

This also explains why the paper keeps a stock Markowitz baseline. If the option-only framework cannot beat ordinary underlying Markowitz after costs and caps, the option structure has not earned its complexity. The stock baseline is not expected to match the option payoff distribution; it is a useful economic hurdle because it asks whether the same underlyings could have been traded more simply. The capped naive option baseline plays the role that $1/N$ plays in the stock literature (DeMiguel et al., 2009): if simple diversification matches

the optimizer after the same costs and caps, the estimation machinery has not paid for itself.

2.2. Premium accounting

Let W_t be portfolio NAV at the rebalance date and let $N_{i,t}$ be the signed number of contracts held in option i . With multiplier m_i , define

$$w_{i,t} = \frac{N_{i,t}m_iC_{i,t}}{W_t}.$$

For equity options $m_i = 100$. For VIX options the multiplier convention is handled in the same way through m_i . The one-period option mark return is

$$r_{i,t+1} = \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}}, \quad \frac{\Delta W_{i,t+1}}{W_t} = w_{i,t}r_{i,t+1}.$$

The multiplier cancels because the weight is already premium-normalized. This is why a cheap out-of-the-money option can have enormous return exposure but still receive a small premium weight. It is also why gross premium, contract concentration, Greek exposure, and stress loss cannot be replaced by a single variance constraint.

Proposition 2.1 (NAV accounting identity). *Let $w_t \in \mathbb{R}^n$ be the option premium-weight vector, r_{t+1} the option mark-return vector, and $r_{f,t}$ the one-period return on residual cash $W_t(1 - \mathbf{1}^\top w_t)$. Ignoring non-option collateral adjustments for notation,*

$$\frac{W_{t+1}}{W_t} - 1 = w_t^\top r_{t+1} + (1 - \mathbf{1}^\top w_t)r_{f,t}, \quad \frac{W_{t+1}}{W_t} - 1 - r_{f,t} = w_t^\top (r_{t+1} - r_{f,t}\mathbf{1}).$$

Proof. The option book is worth $\sum_i N_{i,t}m_iC_{i,t+1} = W_t \sum_i w_{i,t}(1 + r_{i,t+1})$. The residual cash account is worth $W_t(1 - \mathbf{1}^\top w_t)(1 + r_{f,t})$. Adding both accounts, dividing by W_t , and subtracting one gives the identity. $\square$

The empirical tables in this paper use a zero-cash benchmark for comparability across strategy variants. Proposition 2.1 is the conversion rule if a reader wants excess-of-cash returns instead. Positive collateral yield would add income to cash-rich long-option books and raise the hurdle for excess Sharpe ratios. The comparison between the four option variants and the two baselines remains internally consistent because the same convention is applied throughout.

Two details matter for implementation. First, a long option cannot lose more than its premium, but a short option can lose far more than the premium it collected. The premium weight w_i is therefore an accounting weight, not a complete risk measure. Second, every R1 result in this paper is reported for a whole-contract book. The net-utility solver returns continuous premium weights, but each weight is converted to a signed integer number of contracts before the book is scored, so the Sharpe ratios, Sortino ratios, cost figures, and ledgers below all describe integer positions at the standard m_i multiplier. Remark 2.2 states the conversion rule.

Remark 2.2 (Whole-contract rounding). A premium weight w_i implies a continuous contract count $N_i = w_i W_t / (m_i C_{i,t})$. The R1 book truncates N_i toward zero so that integerization

cannot deliberately add risk. It then checks every CVaR, stress, margin, collateral, Greek, and liquidity constraint. If the integer book is infeasible, the decision fails closed to cash. The legacy E1 book rounded to the nearest integer and, wherever a per-contract liquidity cap bound, stepped the magnitude down by one contract so that the realized weight $\hat{w}_i = N_i m_i C_{i,t} / W_t$ never exceeds the cap. All four option books and both capped naive baselines in the legacy tables are scored on those realized integer weights.

2.3. Greek-induced covariance

The option return map comes from a local state expansion. For an equity option on underlying u , write $\Delta S_u = S_{u,t+1} - S_{u,t}$ and $R_u = \Delta S_u / S_{u,t}$. The mark change is decomposed as

$$\Delta C_i = \Delta_{i,t}^{BS} \Delta S_u + \frac{1}{2} \Gamma_{i,t} (\Delta S_u)^2 + \nu_{i,t} \Delta \sigma_{i,t} + \Theta_{i,t} \Delta t + \eta_{i,t+1}.$$

The remainder $\eta_{i,t+1}$ is defined to make this an identity at the rebalance horizon. It absorbs vanna, volga, smile movement, early-exercise premium changes, stale marks, and all higher-order terms not separately modeled. The paper does not require Black–Scholes–Merton to be the true global pricing model for American single-name options. It uses the weaker local fact that option marks can be projected onto spot, curvature, volatility, and time channels (Black and Scholes, 1973; Merton, 1973).

Dividing by $C_{i,t}$ gives premium-normalized state exposure,

$$r_{i,t+1} = b_{i,t}^{\Delta} R_{u,t+1} + b_{i,t}^{\Gamma} R_{u,t+1}^2 + b_{i,t}^{\nu} \Delta \sigma_{i,t} + \theta_{i,t}^* + \epsilon_{i,t+1},$$

where

$$b_{i,t}^{\Delta} = \frac{\Delta_{i,t}^{BS} S_{u,t}}{C_{i,t}}, \quad b_{i,t}^{\Gamma} = \frac{\Gamma_{i,t} S_{u,t}^2}{2C_{i,t}}, \quad b_{i,t}^{\nu} = \frac{\nu_{i,t}}{C_{i,t}}, \quad \theta_{i,t}^* = \frac{\Theta_{i,t} \Delta t}{C_{i,t}}.$$

These coefficients are option-return exposures, not dollar Greeks. Their magnitudes can be large because the option premium can be small. That feature is economically correct: cheap convexity is precisely where leverage, spread costs, and liquidity scarcity enter.

For VIX options the traded anchor is not equity spot but the matched VX futures curve. Using Black’s futures-option logic (Black, 1976), a VIX option j with matched forward $F_{T_j,t}^{VX}$ enters through

$$r_{j,t+1} = b_{j,t}^{\Delta,VX} R_{T_j,t+1}^{VX} + b_{j,t}^{\Gamma,VX} (R_{T_j,t+1}^{VX})^2 + b_{j,t}^{\nu,VIX} \Delta \sigma_{j,t}^{VIX} + \theta_{j,t}^* + \epsilon_{j,t+1}.$$

VIX, VVIX, and term-structure indices are useful state variables, but VX futures and VIX options are the traded volatility instruments. Exact VRO/SOQ settlement is required before VIX option expiry P&L can be promoted from settlement proxy to listed-settlement evidence.

Stacking all contracts gives the conditional factor model

$$r_{O,t+1} = \mu_{O,t} + B_t^G f_{t+1} + \varepsilon_{t+1}.$$

The implemented factor vector contains only underlying returns, centered squared returns, and implied-volatility changes. The fixed matrix B_t^G contains the corresponding premium-

normalized delta, gamma, and vega loadings. Define

$$\Omega_t = \text{Var}_t(f_{t+1}), \quad \Gamma_t = \text{Cov}_t(f_{t+1}, \varepsilon_{t+1}), \quad \Sigma_{\varepsilon,t} = \text{Var}_t(\varepsilon_{t+1}).$$

Because B_t^G is fixed from Greeks rather than estimated as a regression coefficient, there is no reason for Γ_t to vanish.

Proposition 2.3 (Covariance identity). *For any fixed Greek loading matrix B_t^G ,*

$$\text{Var}_t(r_{O,t+1}) = B_t^G \Omega_t (B_t^G)^\top + B_t^G \Gamma_t + \Gamma_t^\top (B_t^G)^\top + \Sigma_{\varepsilon,t}.$$

Equivalently, with $A_t = [B_t^G \ I]$ and $S_t = \text{Var}_t([f_{t+1}^\top, \varepsilon_{t+1}^\top]^\top)$, the option covariance is $A_t S_t A_t^\top \succeq 0$.

Proof. Expand the variance of $B_t^G f_{t+1} + \varepsilon_{t+1}$. The two cross-covariance terms remain. The block form follows by multiplication, and positive semidefiniteness follows because a linear image of a covariance matrix is positive semidefinite. $\square$

If instead $B_t = \text{Cov}_t(r_{O,t+1}, f_{t+1}) \Omega_t^{-1}$, the residual is an OLS projection residual and $\Gamma_t = 0$. That familiar projection formula is a corollary, not the definition used by R1. Empirically, R1 centers aligned training returns and factors, forms $\varepsilon = r - B^G f$, estimates the regularized joint covariance of $[f, \varepsilon]$ on the correlation scale, and transforms it by $[B^G \ I]$. With regularization disabled, the construction reproduces the aligned sample option covariance to numerical precision. Only a final numerical eigenvalue floor is applied after the joint transformation.

2.4. State factors and residual risk

The factor map has a deliberately narrow interpretation. Delta exposure routes option returns into underlying or VX-forward returns. Gamma routes them into centered quadratic movement. Vega routes them into implied-volatility changes. Skew and tail variables proxy where crash insurance is being repriced in the mean and stress models; they are not claimed as covariance loadings.

The residual term is where the model acknowledges its limits. If option marks were perfectly spanned by spot, variance, volatility, skew, and VIX-forward factors, $\Sigma_{\varepsilon,t}$ would be zero. It is not. Residual risk captures early exercise, discrete dividends, surface deformations, quote noise, strike/maturity mismatch, and the fact that a monthly rebalance is not an infinitesimal Ito step. The model can use Greeks without pretending that the Greek map is complete.

The diagonal residual choice in legacy E1 should be read in this spirit. It is not a claim that residual risks are truly independent. It is a regularization decision. In a broad option universe, a dense residual covariance matrix estimated from a short and unbalanced panel can create false diversification and unstable inverse-covariance directions. Diagonal residual covariance keeps contract-specific unexplained risk while refusing to let noisy off-diagonal residual estimates dominate the optimizer. R1 supersedes that approximation with the regularized joint covariance and keeps the observed factor/residual dependence.

2.5. Expected option compensation

Greek covariance alone cannot create a Sharpe ratio. The optimizer needs a conditional mean vector. The economic decomposition is

$$\hat{\mu}_{i,t} = b_{i,t}^{\Delta} \hat{\lambda}_t^S + b_{i,t}^{\Gamma} \hat{\lambda}_t^{var} + b_{i,t}^{\nu} \hat{\lambda}_t^{vol} + b_{i,t}^{skew} \hat{\lambda}_t^{skew} + b_{i,t}^{tail} \hat{\lambda}_t^{tail} + \theta_{i,t}^*.$$

Delta loads on equity compensation. Gamma and theta jointly encode the realized-versus-implied variance wedge. Vega loads on volatility-risk compensation. Skew and tail terms represent compensation for crash insurance and convex volatility exposure, which the option-return literature documents in multiple forms (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009).

The implemented signal is simpler than the general display. It combines carry, a damped delta channel, a vega-routed implied-minus-realized volatility wedge, and skew/tail characteristics. Cheap convexity is treated as systematically expensive rather than anomalous, consistent with the embedded-leverage evidence of Frazzini and Pedersen (2022). The legacy E1 version then removes the raw historical option mean entirely and shrinks the structural forecast toward zero by 0.75. This is not an aesthetic choice. It is the central empirical lesson of the breadth experiments, and it is the option-panel version of the classical breadth-versus-input-quality tension (Grinold and Kahn, 2000): the historical mean is useful when the option universe is small and the training ratio T/n is healthy, but it becomes a ranking-noise generator when the optimizer can choose among many correlated contracts.

Proposition 2.4 (Historical mean error and breadth). *Let the training window contain T monthly observations of n contracts, and let the sample-mean errors $e_i = \hat{\mu}_i^{\text{hist}} - \mu_i$ be mean zero and sub-Gaussian with variance proxies σ_i^2/T , $\bar{\sigma} = \max_i \sigma_i$. Then*

$$\mathbb{E} \left[\max_i |e_i| \right] \leq \bar{\sigma} \sqrt{\frac{2 \log(2n)}{T}},$$

and for any feasible set $\mathcal{C} \subseteq \{w : \|w\|_1 \leq g\}$ the plug-in optimizer's expected objective inflation obeys $\mathbb{E}[\sup_{w \in \mathcal{C}} e^\top w] \leq g \bar{\sigma} \sqrt{2 \log(2n)/T}$. The allocation chosen under the noisy mean therefore sacrifices at most $2g \bar{\sigma} \sqrt{2 \log(2n)/T}$ of true objective value in expectation, and this worst-case loss grows with n at fixed T .

Corollary 2.5 (Bias–variance crossover). *Suppose the structural forecast has a deterministic specification bias bounded by $\beta = \|\mu^{\text{struct}} - \mu\|_\infty$ that does not grow with n . Comparing worst-case objective losses, the historical mean is preferred only when $\bar{\sigma} \sqrt{2 \log(2n)/T} < \beta$, that is, when $n < \frac{1}{2} \exp(T\beta^2/2\bar{\sigma}^2)$. With T fixed, a historical option mean can therefore help at small n and hurt at broad n .*

Proofs are in Appendix A.2. The sub-Gaussian model is a stylization: option panels add correlation, missingness, wide tails, and changing contract composition, all of which raise the effective error at a given (n, T) rather than lower it. The bound therefore understates the practical penalty of ranking many nearly substitutable contracts by noisy estimated means.

The crossover is visible in the artifacts. At eight names, dropping the historical mean hurts the no-VIX gross Sharpe, moving it from 0.842 to 0.725. In the broad universe, keeping

the historical mean is the dominant source of optimizer failure: structural-only forecasts and stronger shrinkage are what allow the 56-name and 57-name books to become viable portfolios. The theory does not say the historical mean is always bad. It says the mean estimator must respect the ratio between sample length and choice set.

Remark 2.6 (Shrinkage to zero as a posterior mean). The 0.75 shrink-to-zero factor has a standard Bayesian reading. If the true premium satisfies $\mu_i \sim N(0, \tau^2)$ and the structural forecast is unbiased with noise $\hat{\mu}_i^{\text{struct}} \mid \mu_i \sim N(\mu_i, s^2)$, the posterior mean is $(1 - \kappa) \hat{\mu}_i^{\text{struct}}$ with $\kappa = s^2 / (s^2 + \tau^2)$. E1's $\kappa = 0.75$ corresponds to a prior signal-to-noise ratio $\tau^2 / s^2 = 1/3$: the forecast is treated as three parts noise to one part signal. Because the structural blend itself carries a weight of 0.75, the effective multiplier on the raw structural signal is $(1 - 0.75) \times 0.75 = 0.1875$. The same regularize-toward-a-simple-target logic is applied to the covariance (Ledoit and Wolf, 2004). Both parameters were fixed in the earlier diagnostic phase and locked before the validation runs; neither is re-tuned inside folds.

2.6. From economic premia to the E1 signal

The structural mean used by E1 can be summarized as four channels. The carry channel is $\theta_{i,t}^*$, the premium-normalized theta contribution under unchanged state variables. The delta channel applies a damped forecast of the underlying's training-window return to $b_{i,t}^\Delta$. The volatility channel routes the implied-minus-realized gap through vega rather than double-counting it through both gamma and theta. The skew/tail channel gives insurance-like wings a negative expected return for long positions and a positive expected return for short positions, subject to stress and liquidity caps.

This construction favors stable economic signs over flexible statistical fit. It does not estimate a separate mean for every contract from that contract's past returns. R1 retains the structural-only, 0.75 shrinkage rule identified during E1 development, but does not describe that choice as prospectively selected.

The no-free-lunch constraint is that structural premia must be paid for in risk budgets. Shorting skew or volatility can earn compensation precisely because those positions lose in crash and volatility-spike states. The optimizer is therefore never allowed to view the mean in isolation. A contract with attractive expected carry can still be excluded by stress loss, vega exposure, beta exposure, margin, concentration, or liquidity.

2.7. Robust conic allocation

Given $\hat{\mu}_t$ and $\hat{\Sigma}_{O,t}$, the headline allocation is

$$\max_{w \in \mathcal{C}_t} \hat{\mu}_t^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma}_{O,t} w - (c_t^+)^\top w^+ - (c_t^-)^\top w^-, \quad (\text{P})$$

where $w^+ = \max(w, 0)$ and $w^- = \max(-w, 0)$. The asymmetric cost vectors include spread, explicit fees, slippage, borrow, and margin funding; charging these costs inside the objective rather than after it follows the cost-aware trading literature (Gârleanu and Pedersen, 2013). The feasible set $\mathcal{C}_t$ imposes premium, Greek, beta/VIX, stress-loss, margin/collateral, concentration, and liquidity constraints. The R1 implementation solves (P) directly. Gross premium is an upper bound, not an equality, so unattractive opportunities produce cash. Capacity enters through hard point-in-time caps rather than a reward for consuming volume.

The policy fixes annualized predicted volatility at no more than 15 percent by choosing $\gamma > 0$ through deterministic training-only bisection. Monthly 95 percent CVaR is at most 10 percent NAV, named stress loss is at most 20 percent, short margin is at most 75 percent, and long premium plus short margin is at most 100 percent NAV. Equity-option shorts flagged for assignment risk are prohibited rather than assigned an ex-post penalty. Because $\gamma > 0$ and the numerical covariance floor makes $\widehat{\Sigma}_{O,t} \succ 0$, the utility objective is strictly concave and selects a unique scale.

The legacy unconstrained maximum-Sharpe geometry remains a useful warning. If $\widehat{\Sigma}_{O,t} \succ 0$ and $\widehat{\mu}_t \neq 0$, the maximum-Sharpe direction is proportional to $\widehat{\Sigma}_{O,t}^{-1} \widehat{\mu}_t$. Unlike a stock-only problem, there is no required $\mathbf{1}^\top w = 1$ budget that bends the efficient frontier into the usual hyperbola.

Proposition 2.7 (Option-only frontier is a ray). *Let $\Sigma \succ 0$ and $\mu \neq 0$. For a target option mean $m > 0$, the minimum-variance premium-weight vector in the unconstrained problem is*

$$w^*(m) = \frac{m \Sigma^{-1} \mu}{\mu^\top \Sigma^{-1} \mu}, \quad \sigma(m) = \frac{m}{\sqrt{\mu^\top \Sigma^{-1} \mu}}.$$

Thus the unconstrained efficient frontier in (σ, m) space is a ray through the origin. This describes why maximum Sharpe cannot choose book size; it is not the R1 allocation rule.

Proof. The Lagrangian for $\min_w w^\top \Sigma w$ subject to $\mu^\top w = m$ yields $2\Sigma w = \lambda \mu$. The constraint pins $\lambda/2 = m/(\mu^\top \Sigma^{-1} \mu)$. Substitution gives the displayed variance. $\square$

The most important operational constraint in this paper is the net liquidity cap

$$|w_i| \leq \bar{w}_{i,t}, \quad \bar{w}_{i,t} = \min \left\{ 0.18, \frac{X \widehat{V}_{i,t}^{train} M_{i,t} 100}{\text{NAV}} \right\}.$$

Here $X = 0.05$, $M_{i,t}$ is the option mark, and $\widehat{V}_{i,t}^{train}$ is a training-window volume statistic. The cap is point-in-time because it uses only train-window volume. It is net because it binds the economic position w_i , not the positive and negative solver legs separately.

Proposition 2.8 (Net caps and representation invariance). *A cap on $|w_i|$ is invariant to algebraic representation of a contract position. A cap on separate long and short legs is not representation-invariant and can create pseudo-cash artifacts.*

Proof. The economic position is $w_i = w_i^+ - w_i^-$. The bound $|w_i| \leq \bar{w}_i$ depends only on that net exposure. Separate bounds on w_i^+ and w_i^- allow both legs to be positive while the net position is small, so a solver can consume displayed capacity in offsetting long/short pairs without taking the intended exposure. $\square$

Proposition 2.9 (Costs and tighter feasible sets). *If costs rise componentwise or $\mathcal{C}_t$ is tightened, the optimal value of (P) weakly decreases. With $\gamma > 0$ and $\widehat{\Sigma}_{O,t} \succ 0$, (P) has at most one optimizer. Zero is feasible, so the program may select an all-cash risky book.*

Proof. For any fixed feasible w , increasing either cost vector weakly lowers the objective, and maximizing over a subset cannot improve the supremum. The negative quadratic is strictly

concave under positive definiteness and $\gamma > 0$, while the remaining terms are concave, giving uniqueness. Feasibility of zero follows from the upper-bound form of every budget. $\square$

These propositions delimit what the empirical sections can and cannot show. The paper is not trying to show that any option signal can be made to work with enough leverage. It asks whether a regularized, cap-constrained option allocation survives when the market frictions that matter for small-account trading are imposed before performance is reported.

2.8. Explicit constraint set

A useful way to read the feasible set is

$$\begin{aligned} \mathcal{C}_t = \{w \in \mathbb{R}^n : & \quad \|w\|_1 \leq g, \quad \sum_{i:w_i < 0} |w_i| \leq s, \quad |w_i| \leq \bar{w}_{i,t} \forall i, \\ & \quad \sum_{i:u(i)=u} |w_i| \leq g_u \forall u, \quad |a_{k,t}^\top w| \leq b_k \forall k, \\ & \quad \ell_{j,t}^\top w \geq -m_j \forall j, \quad \text{CVaR}_{0.95}(-R_t^{\text{train}} w + c_t(w)) \leq 0.10, \\ & \quad \text{ShortMargin}_t(w^-) \leq 0.75, \quad \mathbf{1}^\top w^+ + \text{ShortMargin}_t(w^-) \leq 1, \\ & \quad w_i \geq 0 \quad \text{for assignment-flagged equity options}\}. \end{aligned}$$

The first line controls gross premium, short premium, and contract-level liquidity. The second controls underlying concentration and Greek/beta/VIX budgets. The third controls scenario stress loss, CVaR, collateral, margin, and assignment eligibility. The vectors $a_{k,t}$ include NAV-normalized delta, beta, gamma, equity vega, VIX vega, and other exposure rows. The stress rows $\ell_{j,t}$ reprise the book under adverse spot, volatility, and VIX shocks.

The quadratic risk term is convex in minimization form, and positive/negative parts price asymmetric costs and margin without imposing a gross equality. Liquidity binds the net economic variable. R1 therefore has no split-leg pseudo-cash mechanism: it represents one position vector w , constrains $\|w\|_1$ from above, and leaves unused capital in cash. Appendix A records the legacy maximum-Sharpe geometry only to explain why it was replaced.

2.9. Algorithmic summary

The repaired R1 procedure can be written as a short algorithm.

1. At each monthly decision date, build the eligible option universe and compute premium-normalized Greeks, current costs, marks, volume caps, margin coefficients, assignment flags, and settlement labels.
2. Estimate the regularized joint covariance of delta/gamma/vega factors and Greek residual returns, including both factor/residual cross terms.
3. Form the structural expected-return vector from carry, delta, volatility, skew/tail, and VIX channels; set historical option mean weight to zero; shrink the mean 75% toward zero.
4. Solve net utility with the fixed 15 percent volatility policy and hard CVaR, stress, margin, collateral, assignment, and liquidity limits.

5. Truncate contract counts toward zero, recheck every constraint, fail closed to cash when necessary, and evaluate only the next holding period.

All rolling refits rebuild the representative contracts, mean/covariance estimates, and liquidity caps from the training window. Liquidity caps use training-window volume only. All observations currently available were seen during research development; these checks are internal validation, not an untouched holdout.

3. Empirical application

The empirical application turns the clean conic allocation of Section 2 into an exchange-realistic book by layering three data-driven frictions onto the idealized problem. First, every position is priced through a full transaction-cost stack—spread crossing, fees, slippage, borrow, and margin funding—built from historical market-hours CBBO quotes rather than a flat default spread. Second, each contract carries a volume-aware pre-trade liquidity cap that sizes it to a fixed fraction $X = 0.05$ of its displayed training-window volume, so the optimizer can only take exposure the market actually showed (the cap $\bar{w}_{i,t}$ of Section 2). Third, the continuous premium weights are truncated to whole option contracts at the standard m_i multiplier before any performance number is computed and all constraints are checked again (Remark 2.2). The theory says which exposures to hold; these three layers constrain the book to the spreads, depth, and contract granularity that the historical data actually recorded, and every R1 development result is reported only after all three are imposed. The subsections below give the four universes, the cost stack and its spread sourcing, and the validation protocol in turn.

3.1. Four universes, repaired R1, and legacy E1

The empirical design uses four implementation universes. `orig` is the original eight equity-option names: AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA. `orig+VIX` adds VIX options. `larger` expands the equity-option universe to 56 names. `larger+VIX` adds VIX to that broader set. The four variants are not four different optimizers. R1 applies the same fixed risk policy to each universe.

Config	Universe	Spread input	Status	E1 net	Naive net	Stock Markowitz
<code>orig</code>	8 equities	exact panel CBBO	capacity diagnostic	0.778	0.365	0.909
<code>orig+VIX</code>	8 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.383	-0.890	0.909
<code>larger</code>	56 equities	exact panel CBBO + inferred CBBO proxy	mixed	0.587	0.596	0.909
<code>larger+VIX</code>	56 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.628	0.293	0.909

Table 1: Four empirical scenarios. The stock baseline is ordinary underlying Markowitz. The naive option baseline is the better of capped equal-premium and capped equal-risk option books under the same cost and cap stack.

Legacy E1 used structural-only means, diagonal residual covariance, N -scaled covariance shrinkage, shrinkage-to-zero 0.75, and net liquidity caps. That specification was chosen after the post-2020 performance window had been inspected. Locking it for later CV and Monte Carlo runs prevented fold-by-fold retuning but did not make the earlier window an untouched holdout. R1 keeps the structural mean rule as a development choice, repairs the covariance and objective, and labels every existing observation retrospective.

The contract construction is representative rather than high-frequency. Each rebalance date

maps the eligible option panel into month-end representative contracts and R1 refits using only information available by that decision date before holding the book for one period. This choice makes the backtest auditable: the research question is whether a monthly, capacity-aware option allocation works after costs, not whether intraday execution timing can rescue a fragile signal. It also makes the capacity result sharper. If month-end representative contracts cannot support the desired NAV under a five-percent participation cap, the strategy should not be promoted by assuming invisible intraday depth.

The two baselines are chosen for interpretability. The first is ordinary Markowitz on the underlying stocks. It asks whether option structure adds anything beyond an equity portfolio on the same economic names. The second is a capped naive option book: equal premium or equal risk after applying the same liquidity caps and costs. It asks whether the Greek/mean/covariance theory beats a mechanically diversified options sleeve. The paper's claim requires beating both baselines while remaining deployable under the cap budget.

The two baselines serve different failure modes. Stock Markowitz is a simplicity benchmark: if the option book cannot beat a portfolio of underlying shares, the option engineering may be unnecessary. The capped naive option book is a structure benchmark: if equal-premium or equal-risk option diversification performs as well as R1, then the Greek covariance and structural mean are not adding much. The broad no-VIX result fails this second test in the legacy evidence because its E1 Sharpe is essentially tied with capped naive options.

3.2. Cost stack and spread sourcing

R1 prices the predictable portion of spread crossing, fees, slippage, borrow, and margin funding inside optimization and uses capacity as a hard cap. Legacy E1 deducted the full cost stack only after allocation. The most consequential design decision in that stack is spread sourcing. A flat 10% default spread is not a realistic representation of liquid large-cap options and materially deflates large parts of the option universe. The corrected stack uses exact historical market-hours panel CBBO where available and an inferred CBBO proxy only where matched broad historical quotes are missing.

The full-cost return for a book is therefore not simply gross return minus one spread number. Costs enter through several channels. Spread and fee costs scale with the target option premium traded. Slippage is a further execution allowance. Borrow and margin drag represent the cost of carrying short or collateral-intensive exposure. Assignment penalty is a fail-closed proxy for short American-option exercise risk. Capacity/impact costs penalize positions that try to consume too much displayed volume. The corrected headline results use these costs together; no row is promoted from a gross-only statistic.

Spread sourcing shapes what the results can support because OPRA is the consolidated U.S. options quote and last-sale system,¹ and Cboe sells historical option NBBO snapshots with size, including 15:45 snapshots intended to be more representative than end-of-day liquidity.² Matched broad historical market-hours NBBO/CBBO is generally not free. Therefore this paper separates exact evidence from calibrated execution sensitivity. The baseline eight equity-option rows use exact historical panel CBBO. VIX rows and missing broad-name rows use a point-in-time inferred CBBO proxy calibrated from the liquid CBBO surface. Proxy

¹Options Price Reporting Authority: <https://www.opraplan.com/>.

²Cboe DataShop Option EOD Summary: <https://datashop.cboe.com/option-eod-summary>.

rows are not final execution truth.

Config	Exact CBBO rows	Proxy rows	Proxy share	Median spread
orig	5777	0	0.000	0.019
orig+VIX	5777	536	0.085	0.022
larger	5777	23740	0.804	0.019
larger+VIX	5777	24276	0.808	0.020

Table 2: Spread-source coverage in the corrected cost stack. The proxy share is row-weighted across cost rows, not an exchange-wide liquidity statistic.

The imputation ladder is intentionally conservative in wording even when the calibrated spreads are economically plausible. For the eight equity names, the cost rows are exact historical market-hours CBBO. For `orig+VIX`, the equity-option rows remain exact and the VIX rows are inferred from the liquid-option surface. For broad rows, added names without matched historical CBBO use the same point-in-time proxy. This is enough to replace the stale 10% default in research results, but it is not enough to certify live tradability. The next data upgrade is paid or broker-provided historical market-hours NBBO/CBBO with displayed size matched to every backtest decision date.

The table also explains why the paper gives stronger evidentiary weight to the baseline eight equity rows than to the broad spread rows. `orig` has zero proxy spread rows. `orig+VIX` has exact equity-option rows and proxied VIX rows. The broad variants use proxy rows for most added-name cost observations. That does not invalidate the broad exercise: the proxy is point-in-time and calibrated from observed liquid-option CBBO. It does, however, prevent a final execution claim. The broad result is best read as: under realistic liquid-option spread assumptions, the optimizer survives; under exact historical broad NBBO/CBBO, this still needs to be confirmed.

3.3. Validation protocol

R1 is a monthly walk-forward development exercise. For each return month, the information set, representative contracts, costs, conditional means, joint covariance, margin inputs, and volume caps are reconstructed from observations available no later than the decision date. The resulting integer book is evaluated only in the next period. The four variants use \$1 million NAV and $X = 0.05$ participation. Cash is permitted, so capacity limits reduce deployment instead of inducing pseudo-cash or algebraic infeasibility.

The legacy robustness layer uses 12 chronological groups, 66 CPCV splits run in both a liquid-era (post-2018) design and a claim-window design restricted to the post-2020 evaluation months (Section 5), 78 total CV splits in the PBO summary (Bailey et al., 2017; López de Prado, 2018), one-month purge/embargo, Monte Carlo resampled histories built from circular-block bootstrap draws (Politis and Romano, 1994), Monte Carlo refit stability, repriced synthetic option universes with a historical-cost overlay, circular-block bootstrap path simulations, GARCH/EGARCH-style volatility-path diagnostics (Nelson, 1991) with fallback when the EGARCH sample is too short, drawdown breach simulations, multiple-testing/reality-check inference, and rolling 36-month OOS refits. These checks are designed to attack the result from multiple angles. Because E1 had already been selected using the post-2020 window, these diagnostics do not create a confirmatory holdout.

The purge and embargo rule is simple: one month is dropped around fold boundaries so op-

tion labels whose payoff windows overlap the test block do not leak into training. Lemma A.3 in the appendix records why one month is sufficient: the tested contracts' decision-to-expiry spans are at most about 44 calendar days, while the verifier-measured minimum calendar gap between any usable training month and the tested months—taken on both sides of each test block, since combinatorial folds also train on later months—is 59 days across all folds. The CPCV exercise is distributional, not tradable, because some training sets can include future calendar months relative to a tested block. Rolling OOS is closer to a trading protocol because it refits through time. Monte Carlo refit stability asks whether the procedure selects similar risk-adjusted books under alternate training histories. Repriced synthetic paths ask whether the strategy depends too much on one realized option surface path. Each diagnostic attacks a different source of false confidence.

3.4. Performance metrics and deployability

The primary R1 outcomes are terminal wealth, annualized geometric growth, maximum draw-down, worst month, 95 percent expected shortfall, ruin, margin and collateral breaches, integer feasibility, and capacity. Sharpe and Sortino are secondary descriptors. With monthly returns $R_1, \dots, R_T$, the annualized Sharpe ratio uses

$$\widehat{\text{SR}} = \frac{12 \bar{R}}{\sqrt{12} \hat{\sigma}_R}.$$

The Sortino ratio replaces $\hat{\sigma}_R$ with downside semideviation around the zero-return threshold (Sortino and Price, 1994); drawdown-based and distribution-based complements (Magdon-Ismail and Atiya, 2004; Keating and Shadwick, 2002) appear in the artifact ledgers. Sharpe point estimates on 60 monthly observations carry wide sampling bands (Lo, 2002), which is why Section 5 reports interval and difference-test evidence rather than point estimates alone. These ratios are convenient summaries, but they are not sufficient for a positive verdict. Any realized or reported validation path that reaches zero wealth, breaches margin or collateral, or requires an infeasible integer execution fails the survival gate regardless of Sharpe. This rule directly addresses option-like return smoothing and rare crash losses (Goetzmann et al., 2002).

R1 baseline comparisons use paired circular-block bootstrap draws of the same calendar months. The reported differences are monthly log growth and 95 percent expected shortfall against the matched capped-naive option book and stock Markowitz. This keeps compounding and tail loss central rather than testing only differences in arithmetic Sharpe.

Deployability is a separate requirement. R1 does not require one NAV of gross premium; it reports the exposure that survives the cap, cost, and risk policy and holds the balance in cash. A small deployed book can be a valid defensive decision but cannot be described as evidence for a scalable high-return strategy.

The same logic applies to baseline dominance. A row that is positive but tied with capped naive options does not validate the full theory. It may show that option exposure has a premium, but it does not show that Greek-induced covariance, structural means, and conic constraints improve on simple option diversification. The broad no-VIX row falls into this category.

Universe	Obs.	Mean gross	Terminal wealth	Ann. geom.	Worst month	ES ₉₅	Verdict
orig+VIX	60	0.049	2.814	0.230	-0.062	-0.046	development_survived
larger+VIX	60	0.050	2.401	0.191	-0.059	-0.048	development_survived
orig	60	0.032	1.220	0.041	-0.078	-0.067	development_survived
larger	60	0.038	1.180	0.034	-0.070	-0.061	development_survived

Table 3: R1 monthly walk-forward development summary. Gross is realized premium gross; ES₉₅ is monthly expected shortfall. Any ruin, margin/collateral breach, failed integer repair, or absorbed validation path forces a failed survival verdict.

3.5. Artifact discipline

The empirical claim is backed by generated artifacts rather than hand-entered numbers. The headline figures and tables are rebuilt from machine-readable robustness and cost artifacts. The spread-source coverage table is built from the cost ledger. The validation distributions are built from CPCV, Monte Carlo, refit, and rolling OOS outputs. The paper verifier checks page quality, scope statements, artifact schemas, spread-source policy, VIX settlement handling, and cap point-in-time behavior.

This discipline matters because three design errors identified during the research process would otherwise be invisible in a summary table: charging default spreads to names with observable quotes, penalizing capacity after optimization instead of constraining it before, and a split-leg liquidity formulation that wastes displayed capacity as pseudo-cash. Legacy E1 corrected the first two but retained a gross-equality split representation and omitted costs from its headline objective. R1 corrects all three and versions its artifacts separately so the development history is not erased.

The artifact ledger also protects against a common failure in option research: inadvertently mixing data vintages. Liquidity caps use training-window volume only. Fold-specific models rebuild the contract set and risk estimates from the fold's training window. Broad spread proxy rows are point-in-time calibrated rather than filled from future quote observations. These controls do not make the backtest live trading evidence, but they keep the simulation internally auditable.

4. Results: repaired protocol and legacy development evidence

4.1. R1 retrospective development summary

Table 3 is generated by the repaired monthly walk-forward pipeline. The table leads with deployment and survival quantities rather than Sharpe. Its verdicts are development-sample outcomes only; they cannot be promoted to confirmatory evidence before the prospective 36-month protocol is complete.

All four realized R1 development paths avoid ruin, margin/collateral breach, and failed integer repair. The result comes with materially smaller scale than E1: mean deployed gross is only 3.2–5.0 percent NAV. The two VIX variants have terminal wealth 2.81 and 2.40 and annualized geometric growth 23.0 and 19.1 percent; the no-VIX variants reach 4.1 and 3.4 percent. These are net development returns under the fixed cost model, not proof that the population strategy survives unobserved crashes.

Paired circular-block comparisons sharpen the claim. Against matched capped-naïve options, orig+VIX has a positive monthly log-growth difference with a 90 percent interval of

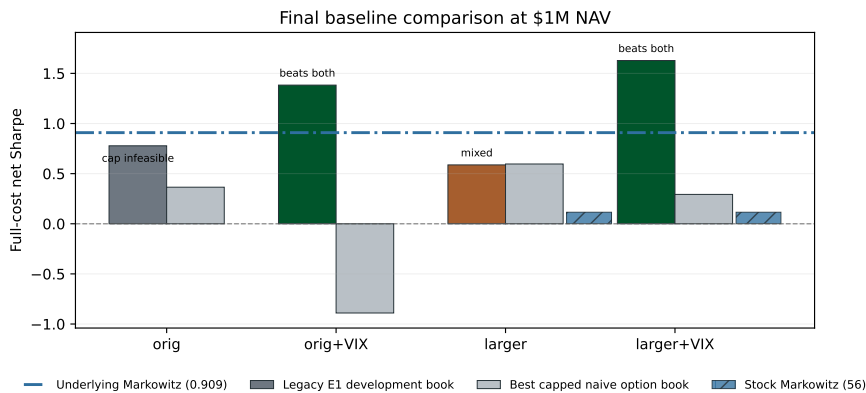


Figure 2: Legacy E1 development comparison at \$1 million NAV. These statistics motivated R1 but do not pass the repaired research-design standard.

Config	Verdict	Gross Sharpe	Net Sharpe	Net Sortino	Naive Sharpe	Stock Sharpe	Rolling Sharpe	MC resample p05	MC refit p05
orig	capacity diagnostic	0.975	0.778	1.783	0.365	0.909	0.585	0.199	0.570
orig+VIX	pass	1.675	1.383	3.551	-0.890	0.909	1.033	0.873	1.178
larger	mixed	0.847	0.587	1.298	0.596	0.909	0.480	-0.001	0.537
larger+VIX	pass	2.010	1.628	4.377	0.293	0.909	1.273	1.096	1.360

Table 4: Legacy E1 numeric scoreboard behind Figure 2. Returns are net of ex-post costs unless a column is explicitly labeled gross.

0.063 to 0.256, while `larger+VIX`'s interval is -0.001 to 0.115 and therefore does not exclude zero. Against stock Markowitz, `larger+VIX` has a positive interval of 0.005 to 0.023 ; `orig+VIX`'s interval includes zero. The no-VIX comparisons are mixed. Thus R1 supplies a safer, smaller development book, not a universal dominance result.

4.2. Legacy E1 comparison

The following E1 numbers are retained to document why the repair was undertaken. They were computed after the post-2020 window had informed specification choice, costs were deducted after allocation, and the maximum-Sharpe split formulation could leave gross undeployed through offsetting algebraic legs. They are retrospective development statistics, not R1 headline results and not confirmatory evidence. Within that legacy exercise the point estimates concentrate in the VIX-enabled universes. `larger+VIX` is the strongest candidate, with net Sharpe 1.628 and net Sortino 4.377. `orig+VIX` also beats both baselines, with net Sharpe 1.383 and net Sortino 3.551. `larger` without VIX is positive but essentially tied with the capped naive option book and below stock Markowitz. `orig` without VIX has positive point estimates but fails the deployability test because the hard liquidity-cap budget is below one NAV.

Gross and net results are reported side by side so the reader can see exactly what the cost stack removes. The gross-to-net Sharpe pairs are $0.975 \rightarrow 0.778$ for `orig`, $1.675 \rightarrow 1.383$ for `orig+VIX`, $0.847 \rightarrow 0.587$ for `larger`, and $2.010 \rightarrow 1.628$ for `larger+VIX`. The corrected cost stack therefore removes between 0.20 and 0.38 Sharpe points across the four books. This verifies net return accounting, not cost-aware selection.

Two comparisons matter. Against underlying stock Markowitz, the VIX-enabled option

books earn a higher net Sharpe in this simulation while still being subject to explicit option costs and liquidity caps. Against capped naive options, the VIX-enabled E1 books earn a large optimizer edge: `orig+VIX` moves from a negative naive option Sharpe to a positive E1 Sharpe, and `larger+VIX` beats the capped naive option benchmark by more than one Sharpe point. The no-VIX broad row does not earn this interpretation because 0.587 for the optimizer versus 0.596 for capped naive is not an economically meaningful edge in either direction.

The Sortino ratios tell the same story from downside variation. `larger+VIX` reaches net Sortino 4.377, and `orig+VIX` reaches 3.551. The no-VIX rows are materially weaker: `orig` has Sortino 1.783 but fails capacity, while `larger` has Sortino 1.298 and does not beat the stock baseline. The downside metrics therefore reinforce the ranking rather than changing it.

It is also important that the result is not driven by treating relaxed or infeasible rows as deployable candidates. A positive point estimate is not sufficient. The candidate must survive the hard net liquidity cap, full-cost accounting, and baseline comparisons. This is why the original no-VIX row is not promoted even though its E1 Sharpe is positive, and why the broad no-VIX row is marked mixed even though it is deployable.

4.3. Mechanism: breadth, mean regularization, and caps

The result is not simply that more names improve the opportunity set. Breadth helps only after the estimator and liquidity formulation are fixed together. The dominant estimator mistake was the historical mean. In the `larger+VIX` universe, dropping it and using structural-only forecasts recovers much of the gross Sharpe that the broad paper configuration lost. Diagonal residual covariance and N -scaled shrinkage then remove a second overfitting channel. The development-selected combination reaches a gross Sharpe of 2.010 for the broad VIX-enabled book and produces the net Sharpe 1.628 after the corrected cost stack.

Arm	orig	orig+VIX	larger	larger+VIX
Full E1	0.778	1.383	0.587	1.628
No carry	0.774	0.463	0.574	0.483
No delta	0.370	1.479	0.672	1.684
No vol/VRP	0.772	1.382	0.587	1.600
No skew/tail	0.772	1.372	0.586	1.629
No relative value	0.822	1.355	0.504	1.570

Table 5: Legacy structural-mean channel ablation. Each row re-solves and re-scores E1 book with one expected-return channel set to zero, holding the covariance, caps, and cost stack fixed; cells are full-cost net Sharpe. The Full E1 row reproduces the headline scoreboard by construction and is enforced by the verifier.

Table 5 shows which expected-return channel actually carries each book. The answer is not uniform across universes. For the two VIX-enabled books the carry channel is load-bearing: removing it collapses `orig+VIX` from 1.383 to 0.463 and `larger+VIX` from 1.628 to 0.483, while removing the volatility, skew/tail, or relative-value channels moves those books by at most a few hundredths. The optimizer's VIX edge in this sample is therefore priced premium capture under unchanged state variables, disciplined by the stress and vega budgets, rather than a directional volatility forecast. For the no-VIX eight-name book the ranking reverses: the delta channel is the engine (0.778 falls to 0.370 without it), and carry is nearly irrelevant.

At broad no-VIX breadth the relative-value term contributes meaningfully (0.587 versus 0.504 without it).

Two readings of this table are excluded. First, the arms in which a removed channel slightly raises the net Sharpe (dropping delta lifts `orig+VIX` to 1.479 and `larger+VIX` to 1.684) are not promoted as improved candidates: E1 was locked before the validation runs, these are post-hoc diagnostics of the locked object, and differences of this size sit well inside the sampling bands reported in Section 5. Second, the ablation does not claim channel-level attribution of realized P&L; it measures how the selected book changes when the optimizer stops seeing one premium source.

The liquidity cap matters for a different reason. A post-hoc capacity penalty lets the optimizer select positions first and then punishes them after the fact. A hard pre-trade cap changes the feasible set before selection. The net-cap formulation also fixed a real solver pathology: when caps were applied to split long/short legs, the solver could burn offsetting exposure as pseudo-cash and consume good contracts' capacity. Binding $|w_i|$ after netting removed that artifact and left non-degenerate cases unchanged.

The VIX result is economically plausible. Equity options provide single-name convexity, skew, and carry exposure. VIX options add a traded volatility instrument (Whaley, 1993) that can respond directly to market-wide volatility jumps. The empirical result says that this extra state channel matters in the tested sample. It does not say that VIX options are free hedges: they have their own spreads, settlement requirements, and wing-liquidity caveats, and the paper keeps those rows marked as proxied where matched historical CBBO is absent.

The cap result is equally important. The old optimizer could generate attractive gross allocations that were later destroyed by capacity costs. The new formulation moves the constraint into the decision. That changes the selected book, not just the accounting afterward. At \$1 million NAV and $X = 0.05$, the optimizer chooses a smaller, cleaner set of trades that the cap vector can support. At larger NAV, the cap vector itself becomes too small to support full deployment, so the correct conclusion is not to increase leverage but to stop promoting the book.

Config	Candidates	Active	Top 5 share	Deployed gross	At cap share	Cap budget
orig	49	47	0.421	0.497	0.064	0.739
orig+VIX	54	54	0.324	0.723	0.093	1.093
larger	262	243	0.233	0.594	0.078	1.160
larger+VIX	267	253	0.296	0.836	0.083	1.515

Table 6: Locked E1 book shape at \$1 million NAV and $X = 0.05$. Candidates is the number of eligible contracts, Active the number with nonzero weight, Top 5 share the premium share of the five largest positions, At cap share the fraction of active positions at their liquidity cap, and Cap budget the sum of per-contract caps in NAV units. Deployed gross is cross-checked against the locked robustness ledger by the verifier.

Table 6 quantifies how binding this discipline is. In every universe the optimizer touches nearly every eligible contract—all 54 in `orig+VIX`, and 47 of 49, 243 of 262, and 253 of 267 in the others, the remainder rounding to zero whole contracts. Because whole-contract rounding steps each cap-binding leg just inside its cap, realized utilization clusters near but rarely exactly at the cap: median leg utilization is 0.86–0.94 across the four books, while

only 6–9 percent of legs land within one percent of the cap after rounding. Breadth buys diversification of the cap budget rather than concentrated conviction: moving from 49 to 267 candidates lowers the top-five premium share from 0.421 to 0.296 while raising the deployable cap budget from 0.739 to 1.515 NAV units. The book’s shape is set jointly by the cap vector and the mean/covariance inputs, not by the covariance alone. The `orig` row also makes the capacity diagnosis concrete: its cap budget of 0.739 NAV is below one, which is exactly why that row is labeled capacity-infeasible despite its positive Sharpe.

4.4. Interpretation by scenario

The `orig` no-VIX row is useful mainly as a control. It uses exact historical CBBO for the eight equity names and achieves a positive point estimate, but it is capacity-infeasible under the hard cap budget. The result therefore validates pieces of the framework without becoming a deployable headline book.

The `orig+VIX` row is a stronger result. It uses exact equity-option CBBO for the eight names and an inferred proxy for the VIX spread rows. It beats stock Markowitz and capped naive options, and its MC refit p05 remains above one Sharpe point. The caveat is execution data for VIX rows, not the basic allocation mechanism.

The `larger` no-VIX row is mixed. The regularization and cap fixes make the result positive, but the optimizer does not beat capped naive options by a meaningful amount and it remains below the stock Markowitz Sharpe. This row is evidence that breadth alone is not enough.

The `larger+VIX` row is the strongest result. It has the best E1 Sharpe, best Sortino, strongest rolling Sharpe, and strongest refit-stability statistics. It also carries the largest spread-imputation dependence, because roughly four-fifths of broad cost rows use the inferred CBBO proxy rather than exact historical CBBO; its performance should be read under the calibrated spread assumptions of Section 3.

4.5. Economic interpretation

The four scenarios separate three mechanisms. The first is option structure: E1 has to beat capped naive options, otherwise the theory adds little. The second is volatility instrument access: adding VIX materially improves both the original and broad universes. The third is breadth: adding names helps only when the estimator is regularized and VIX is present. Without VIX, the broad universe becomes positive but does not clear the baseline hurdle.

The result also shows why option-only strategies should be evaluated by deployable notional rather than by nominal contract count. The broad universe contains many more contracts, but the added liquidity budget is much smaller than the added choice set. That asymmetry creates the central estimation-capacity tradeoff. More names reduce idiosyncratic dependence and create more spread opportunities, but they also increase mean-estimation noise and may not increase executable premium capacity enough.

The strongest practical lesson is that breadth and liquidity caps are complements. A breadth expansion without cap discipline lets the optimizer choose thin options and later pay capacity penalties. A cap without mean regularization can keep trades small but still select noisy expected returns. The positive VIX-enabled rows appear only when both pieces are present.

4.6. What the point estimates do and do not say

The point estimates are economically meaningful because they are full-cost net numbers, but they remain sample estimates. A net Sharpe of 1.628 for larger+VIX does not mean the population Sharpe is known to three decimals. It means that, in the realized development path and under the corrected ex-post cost stack, the legacy E1 book earned a substantially better risk-adjusted return than both baselines. The validation section then asks whether that ordering survives alternate histories and refits, and Table 9 attaches confidence intervals, deflated Sharpe ratios, and formal Sharpe-difference tests to these point estimates.

The large Sortino ratios should be read similarly. Option books can have asymmetric return distributions, so downside semideviation is a useful complement to standard deviation. But Sortino ratios can also look high when the sample has few downside months. That is why the paper does not rely on Sortino alone. It reports rolling OOS, MC lower tails, drawdown breach rates, and CPCV stress tests in the same result package.

The stock Markowitz baseline is not a perfect substitute for the option book because it lacks convexity and volatility exposure. It is nevertheless a hard economic comparison: if an option strategy cannot beat a simple stock portfolio after paying option trading costs, the option-specific complexity is not justified. The capped naive option baseline is the complementary comparison: it keeps the option asset class but removes the theory. Passing both baselines is why the VIX-enabled rows deserve attention.

The broad no-VIX row is the clearest example of why this distinction matters. It has a positive Sharpe, so it is not a failed strategy in the narrow sense. But it is almost identical to capped naive options and below the stock baseline. A paper that promoted that row as evidence for Greek Markowitz would overstate the finding. Here it is marked mixed and used to show that breadth without a traded volatility instrument is not enough in the current sample.

5. Robustness and distributional checks

5.1. Cross-validation and overfitting controls

The validation layer first asks whether the result depends too much on one chronological split. Combinatorially purged cross-validation is run in two designs. The *liquid-era* design divides the post-2018 monthly panel into 12 contiguous groups; each fold refits the legacy E1 procedure on the remaining groups after a one-month purge and embargo, scores the held-out pair, and the $\binom{12}{2} = 66$ splits recombine into 11 complete backtest paths. The *claim-window* design applies the identical machinery but partitions only the post-2020 evaluation months on which every claim in this paper is made, training each fold on all earlier history plus the non-held-out claim months. The liquid-era restriction is deliberate, and the next paragraphs justify it directly from the data: recombining folds across the entire 2015–2026 panel fails for every book, but the failure is concentrated in 2015–2017 months when the listed-options panel was too thin for the modern per-contract cost model to be realistic, not in the COVID period. The PBO summary combines the 12 blocked splits and 66 CPCV splits, for 78 total CV splits.

CPCV of either design is a robustness diagnostic, not a tradable out-of-sample history. Some CPCV training sets include months after the test months, by construction. Its role is to stress the selection and path distribution, estimate probability of backtest overfitting, and

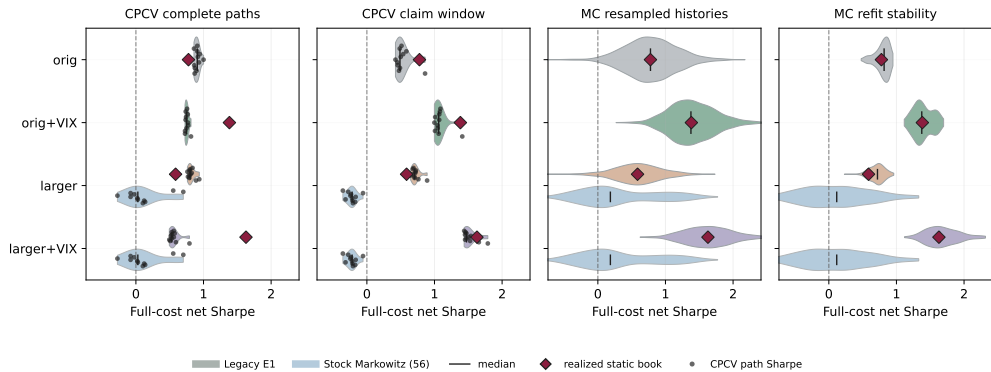


Figure 3: Distributional validation for all four variants. Each shape is the density of full-cost net Sharpe ratios across validation paths (CPCV panels overlay the 11 individual path values as points); black bars mark medians and diamonds mark the realized static-book Sharpe. MC refit and resampled evidence supports the VIX-enabled candidates; liquid-era and claim-window CPCV net paths are positive for all four variants, and the broad universes add a zero-cost 56-equity stock Markowitz overlay. Under the repaired survival rule, any absorbed-zero path fails regardless of these Sharpe distributions.

Config	Liquid net p05	Liquid net p50	Default share	Claim net p05	Claim net p50	Rel liquid p05	Rel claim p05
orig	0.860	0.908	0.000	0.437	0.490	-0.561	-0.125
orig+VIX	0.727	0.749	1.000	1.003	1.062	0.214	2.257
larger	0.766	0.802	0.000	0.660	0.706	-0.265	-0.152
larger+VIX	0.500	0.532	1.000	1.445	1.474	0.103	1.501

Table 7: CPCV path distributions under the two designs. Liquid columns are recombined-path net Sharpe quantiles when folds span the post-2018 liquid-panel era; Default share is the fraction of those net paths whose compounded wealth is absorbed at zero, reported separately from the Sharpe statistics (it is driven, for the VIX books, by the single March-2020 volatility-spike month that recombination places in every path). Claim columns restrict the held-out groups to the post-2020 evaluation window. Rel columns are fifth percentiles of per-path Sharpe differences between E1 and the matched capped-naïve book on the same folds and costs, under the liquid-era and claim-window designs respectively.

reveal whether a result comes from one lucky chronological window. The paper therefore reports CPCV and PBO as validation evidence, while the trading claim remains tied to legacy E1 and the static/rolling internal-validation protocols.

Restricted to the liquid 2018–2026 era, the recombined CPCV net-Sharpe distributions are positive for every variant—fifth-percentile and median values of 0.86/0.91 for *orig*, 0.73/0.75 for *orig+VIX*, 0.77/0.80 for *larger*, and 0.50/0.53 for *larger+VIX* (Table 7), on gross paths that are solidly positive (median 0.98–1.33). The reason the full 2015–2026 CPCV fails is specific, and it is not COVID. Recombined paths force fold-refit books through 2015–2017 months in which the corrected \$1 million cost stack is punitive on a thin early listed-options panel, producing isolated months with modeled net returns below -100% —a cost artifact, not an economic outcome. Those pre-2018 months, not 2020, drive the failure: excluding calendar 2020 alone leaves the full-sample fifth percentile essentially unchanged (-0.454 to -0.455 for *larger+VIX*), whereas dropping the pre-2018 months flips every book positive.³ One genuine tail does survive the restriction. The VIX-enabled books still carry

³The full 2015–2026 CPCV ledger is superseded in the current artifact set by the liquid-era design; the full-sample and ex-2020 quantiles quoted here trace to the archived pre-restriction path ledger in the repository history.

a single March-2020 volatility-spike month whose modeled loss exceeds a month of premium, and because CPCV recombination places that month in every path, the compounded VIX wealth paths are absorbed at zero (Default share 1.0) even though the return-based path Sharpe stays positive; that is a disclosed short-volatility tail, not a data artifact. Under R1's hard survival gate, this evidence forces a failed verdict for the legacy VIX candidates; positive Sharpe cannot override economic insolvency. The claim-window design, on the pure post-2020 period, confirms the reading, with fifth-percentile and median path Sharpes of 0.44/0.49 for orig, 1.00/1.06 for orig+VIX, 0.66/0.71 for larger, and 1.45/1.47 for larger+VIX.

The relative statistic aligns with the paper's verdicts. Because the naive baselines are scored on the same folds under the same costs, per-path Sharpe differences between E1 and the matched capped-naive book difference out the common cost drag. On the liquid-era folds, E1 beats its matched naive book at the fifth percentile for the two VIX-enabled universes (+0.21 and +0.10) but not for the two no-VIX universes (−0.56 and −0.27); the claim-window relative paths separate the same way, more sharply, at +2.26 and +1.50 for the VIX books and −0.13 and −0.15 for the no-VIX books. The no-VIX shortfall is worth stating carefully: the locked static orig book does beat its naive baseline (Table 9), but that advantage is not stable under fold-level refitting, which is the CPCV analogue of the mixed and capacity-diagnostic verdicts those rows already carry.

Two features of this analysis bear emphasis. First, the pass/mixed verdicts reported in Section 4 use the liquid-era CPCV tail as one of five robustness criteria, but a book that clears all five robustness screens without beating its capped-naive baseline is still reported as mixed rather than promoted; this is why the broad no-VIX book remains mixed even though its liquid-era CPCV is positive. Second, the full 2015–2026 result is not quietly dropped because it fails: the failure, and its mechanism—the modern cost stack applied to a pre-2018 thin panel, with 2020 shown not to be the cause—are reported here directly, so restricting the long-history CPCV to the liquid era is a disclosed data-quality decision rather than a search for a friendlier statistic.

Traditional Markowitz on the broad universe is a revealing baseline under the same screens. A zero-cost 56-equity tangency portfolio—the textbook plug-in Markowitz rule, given every advantage by charging it no trading cost—earns only 0.115 net Sharpe over the test window, and its recombined and resampled distributions are weak throughout: liquid-era CPCV fifth percentile −0.18, claim-window −0.34, Monte Carlo resampled −0.46, and Monte Carlo refit −0.64. The estimation error that an eight-name stock Markowitz book largely avoids (Sharpe 0.91) overwhelms the plug-in rule at 56 names, which is exactly the breadth-versus-estimation tension the option framework answers with structural means and regularized covariance. The broad larger+VIX E1 book beats this broad stock Markowitz baseline on every one of these screens (Figures 2 and 3), and the broad no-VIX larger book, though it only ties its capped-naive option baseline, still clears the broad stock baseline comfortably.

The PBO interpretation is similarly bounded. The PBO rank screen asks whether the row chosen in sample tends to rank poorly out of sample across the cross-validation splits. Passing this screen reduces concern that the observed row is merely a backtest winner among many alternatives. It does not eliminate development risk, because the research program itself learned from earlier failed configurations. The paper handles that by locking E1 before the final robustness run and by reporting the no-VIX and proxy-spread caveats plainly.

Robustness summary by test

orig	diagnostic	fail	pass	pass	pass	pass	pass	pass	mixed
orig+VIX	pass	pass	pass	pass	pass	pass	pass	pass	mixed
larger	mixed	pass	pass	pass	mixed	pass	pass	mixed	fail
larger+VIX	pass	pass	pass	pass	pass	pass	pass	mixed	fail
	Baselines	Capacity	CPCV net (2018+)	CPCV net claim	MC resampled	MC refit	Rolling OOS	PBO rank	Repriced overlay

Figure 4: Robustness summary by test. Repriced synthetic paths use a historical-cost overlay rather than synthetic NBBO; the repriced no-premium overlay is now the binding stress, while both the liquid-era and claim-window CPCV are positive for all four variants.

5.2. Monte Carlo, repricing, and path risk

The Monte Carlo layer has three roles. Resampled historical universes preserve the joint cross-sectional strategy-return dependence while changing month order through circular-block sampling. Refit stability resamples training histories, refits the locked procedure, and measures how sensitive the final book is to training-sample variation. Repriced synthetic universes rebuild option paths from simulated state variables and model prices, then apply a documented historical-cost overlay.

The repriced layer is a stress world by construction. It is not synthetic NBBO. Black–Scholes and Black–76 repricing can test exposure to spot, volatility, VIX, and state-path assumptions, but it does not reproduce historical order books, queue priority, displayed size, or market-maker widening. Repriced synthetic paths therefore answer a model-risk question, not an execution-quality question. Tail-path simulation diagnostics are path-risk diagnostics only.

The failure or mixed status of repriced overlay rows should not be over-read. A repriced Black–Scholes/Black–76 world does not contain the same historical option risk premia as the observed market. If the strategy earns compensation from selling or buying specific real-world insurance premia, a synthetic repricing model can remove part of the edge by construction. The useful information is sensitivity: which variants remain stable when the option surface is rebuilt from state paths, and which variants depend heavily on the observed realized surface.

Drawdown breach simulations and GARCH/EGARCH-style volatility diagnostics then ask a different question: how likely is the strategy to violate a risk tolerance under stressed path dynamics? EGARCH is used when the sample is long enough; otherwise the implementation falls back to the simpler volatility-path diagnostic. The paper does not promote those simulations to alternate realized histories. They are scenario tools for risk review.

Config	Baselines	Capacity	CPCV net (2018+)	CPCV net claim	MC resampled	MC refit	Rolling OOS	PBO rank	Repriced overlay
orig	diagnostic	fail	pass	pass	pass	pass	pass	pass	mixed
orig+VIX	pass	pass	pass	pass	pass	pass	pass	pass	mixed
larger	mixed	pass	pass	pass	mixed	pass	pass	mixed	fail
larger+VIX	pass	pass	pass	pass	pass	pass	pass	mixed	fail

Table 8: Pass/mixed/fail summary by robustness test. “PBO rank” is based on the within-configuration full-cost-net PBO row.

The robustness matrix supports a narrow conclusion. The VIX-enabled rows pass the baseline, capacity, liquid-era CPCV, claim-window CPCV, MC resampled, MC refit, and rolling

OOS screens; on the PBO-rank screen *orig*+VIX passes while *larger*+VIX is mixed, and the most conservative repriced no-premium overlay is the one stress neither clears (mixed for *orig*+VIX, fail for *larger*+VIX). The broad no-VIX row is mixed because its realized point estimate is positive but its optimizer edge over capped naive is negligible; its liquid-era CPCV is positive but its resampled lower tail crosses zero and it does not beat either baseline. The original no-VIX row remains a diagnostic because capacity, not forecast quality, blocks a deployable verdict.

The distributional checks therefore support a ranking rather than a binary sales claim. The first tier is *larger*+VIX, which has the strongest point estimate and the best refit/rolling support. The second tier is *orig*+VIX, which is cleaner on equity spread sourcing but still uses proxied VIX spreads. The third tier is *larger*, which is positive but not meaningfully better than capped naive or stock Markowitz. The fourth tier is *orig*, which is useful for exact-CBBO diagnostics but not deployable under the cap budget.

5.3. Multiple testing and rolling OOS

The multiple-testing layer prevents the paper from treating the best observed row as a free discovery. The reality-check and deflated-Sharpe style diagnostics (White, 2000; Hansen, 2005; Bailey and López de Prado, 2014) account for strategy search and finite-sample selection; HAC-based family inference (Newey and West, 1987) stays in the artifact ledgers. The legacy E1 robustness run reduces fold-level retuning but not this caveat: the final validation does not re-tune the specification inside test folds. Nevertheless, the model was developed through earlier diagnostics, so the paper reports the remaining selection risk rather than claiming a pristine randomized experiment.

Config	Net Sharpe	CI lo	CI hi	PSR	DSR	dSR stock	p stock	dSR naive	p naive
<i>orig</i>	0.778	0.163	1.433	0.973	0.945	-0.131	0.776	0.413	0.014
<i>orig</i> +VIX	1.383	0.895	1.971	1.000	0.997	0.474	0.383	2.273	0.000
<i>larger</i>	0.587	0.029	1.173	0.925	0.745	-0.322	0.436	-0.009	0.967
<i>larger</i> +VIX	1.628	1.111	2.245	1.000	0.996	0.719	0.179	1.335	0.000

Table 9: Formal inference on the static full-cost net books over the 60 test months. CI lo/hi are 90 percent circular-block bootstrap bands for the net Sharpe. PSR is the probabilistic Sharpe ratio against a zero benchmark and DSR is the legacy deflated Sharpe ratio charged for 22 recorded grid candidates. The repaired research-trial registry shows that this is only a lower subset of the broader development path. dSR columns are annualized Sharpe differences of E1 minus the stock Markowitz baseline and minus the matched capped-naive baseline, with two-sided Jobson–Korkie–Mennel p-values.

Table 9 states what the sample can and cannot support. The two VIX-enabled books are individually solid: their 90 percent Sharpe bands exclude zero (0.895–1.971 for *orig*+VIX, 1.111–2.245 for *larger*+VIX), their probabilistic Sharpe ratios both round to 1.000, and their legacy deflated Sharpe ratios are 0.997 and 0.996 after charging for 22 recorded candidates (Bailey and López de Prado, 2014). Earlier iterations cannot all be reconstructed, so these values are not presented as a complete selection correction. The no-VIX rows are weaker on the same evidence: the *larger* band touches zero at its lower edge and its DSR falls to 0.745.

The Sharpe-difference tests (Jobson and Korkie, 1981; Mennel, 2003), computed on paired monthly series and confirmed by a paired circular-block bootstrap in the spirit of

Ledoit and Wolf (2008), draw the sharpest distinction in the paper. Against the capped-naive baselines, the optimizer edge of the VIX-enabled books is large and statistically significant: +2.273 Sharpe points for `orig+VIX` and +1.335 for `larger+VIX`, both with $p < 0.001$, and +0.413 with $p = 0.014$ even for the capacity-infeasible `orig` diagnostic. The `larger` no-VIX edge over naive is essentially zero and slightly negative (-0.009 , $p = 0.967$), which is the formal version of the “mixed” verdict. Against stock Markowitz, however, no variant’s Sharpe advantage is statistically significant on 60 months: the differences range from -0.322 to $+0.719$ with p -values between 0.179 and 0.776.

This asymmetry belongs in the statement of results, not in a footnote. The finding that survives formal testing is that Greek-induced allocation with VIX access beats naive option diversification after full costs. The finding that the VIX-enabled books beat the stock baseline is a point-estimate ordering that the current sample length cannot separate statistically; a 0.72 annualized Sharpe gap simply requires more than 60 monthly observations to resolve at conventional levels. The paper therefore leans on the baseline comparisons, the rolling OOS refits, and the MC refit distributions jointly rather than on any single test.

Rolling 36-month OOS refits are the most trading-like validation in the current package. They refit the model through time and score subsequent months under the same cost and liquidity rules. The rolling Sharpe values are 1.273 for `larger+VIX`, 1.033 for `orig+VIX`, 0.585 for `orig`, and 0.480 for `larger`. Those numbers reinforce the ordering in the static scoreboard: VIX-enabled E1 performs; no-VIX breadth is not enough.

Figure 5 is the most useful visual bridge from the scoreboard to the validation logic. It shows that the VIX-enabled E1 rows do not win only through one terminal statistic; their cumulative paths recover from the early sample drawdown and then separate after 2024. It also shows why the capped-naive VIX baselines are not acceptable substitutes: the matched naive VIX books decay sharply even though some naive no-VIX paths recover later. The regular stock Markowitz path is a strong baseline, but it does not match the rolling OOS compounding of the two VIX-enabled E1 variants.

5.4. Interpreting the robustness tests

The tests are not averaged into a single meta-score because they answer different questions. Capacity asks whether the target book can be deployed at the specified NAV. Baseline dominance asks whether the option theory beats simpler alternatives. Liquid-era CPCV asks whether path ordering and fold composition create fragility beyond the claimed period within the liquid panel; claim-window CPCV asks the same question inside it. MC resampling asks whether the realized sequence is unusually favorable. MC refit asks whether the procedure is stable to training histories. Repriced overlay asks whether the result depends on the realized option surface and cost path. Rolling OOS asks whether the model can be refit through time without relying on one static split.

The pass/mixed/fail labels should therefore be read qualitatively. A repriced no-premium overlay failure does not automatically reject the static candidate, because that design is a deliberately adversarial cost-and-surface stress rather than a trading protocol. A capacity failure does reject deployable status, because a book that cannot be formed at the target NAV is not a trading candidate. A repriced overlay failure is a model-risk warning, not proof that historical edge is illusory. The paper’s final ranking follows the tests that are closest to the intended

claim: deployability, full-cost baseline dominance, claim-window CPCV, MC refit stability, rolling OOS, and spread-source transparency.

This framework is also why the paper does not chase a perfect all-green heatmap. A stress battery worth reporting should contain visible failures. If every robustness test were passed easily, the likely explanation would be that the tests are too weak. The goal is to identify which limitations remain after the main methodological errors have been fixed.

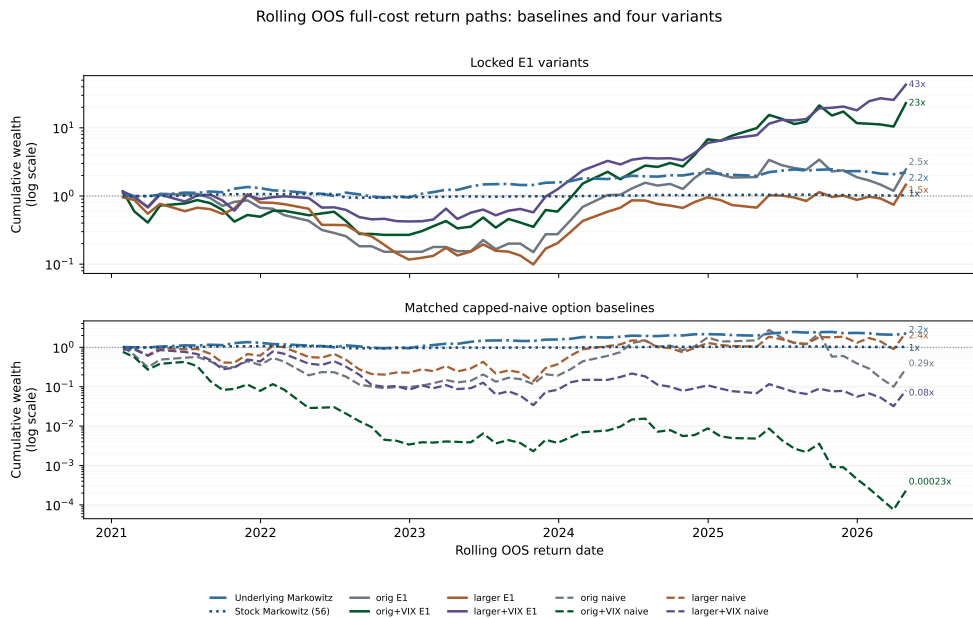


Figure 5: Rolling OOS full-cost cumulative wealth paths. The top panel compares the four legacy E1 variants with underlying stock Markowitz. The bottom panel compares the matched capped-naïve option baselines with the same stock baseline. The VIX-enabled E1 variants are the only option books that both compound strongly and beat the stock baseline in this walk-forward ledger.

6. Capacity, execution, and limitations

6.1. Capacity and scale

The main practical brake is market depth. At \$1 million NAV, the legacy VIX-enabled books passed E1's capacity screen but fail R1's survival interpretation because of the absorbed-zero paths. Above roughly \$2 million, the representative month-end option contracts no longer support full deployment under the tested participation caps. At \$5 million and above, the sum of train-window liquidity caps falls below one NAV for every tested configuration and participation level. Positive Sharpe rows that deploy only a tiny fraction of NAV are economically vacuous and are not promoted.

Breadth increases the cap budget, but not in proportion to the number of underlyings. Moving from eight equity names to 56 names increases the number of candidate contracts by far more than it increases executable month-end premium capacity. Many added-name options are thin. The option market, not the optimizer, becomes the wall; this is a capacity diagnostic rather than a positive trading claim.

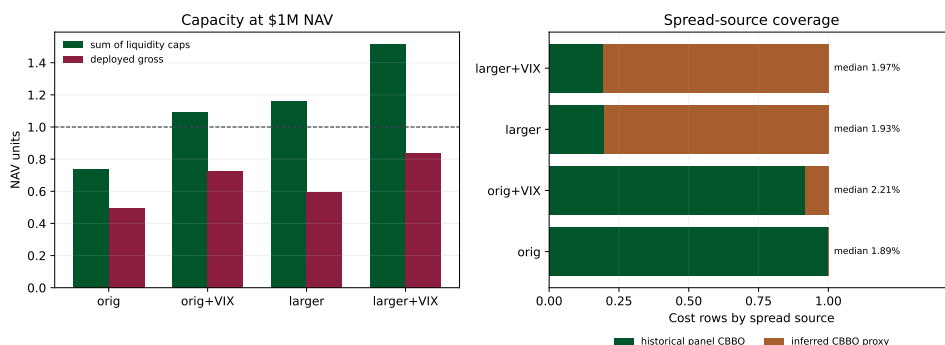


Figure 6: Capacity and spread-source coverage. The left panel shows that deployability is a binding part of the result. The right panel separates exact historical panel CBBO from inferred CBBO proxy rows.

This finding also determines how the capacity result should be read for implementation. Sizing should not begin by asking how much capital can be allocated to the best Sharpe row; it should ask what NAV can be deployed while respecting per-contract displayed size, per-underlying concentration, and margin limits. The research answer points to a small-account allocation, not a scalable standalone fund. Additional capital is better spent expanding the execution universe with verified quotes and displayed size than multiplying the same representative contracts.

6.2. Execution caveat

The spread proxy is the largest remaining empirical caveat. It is a major improvement over the old default spread assumptions and stale quote snapshots, but it still cannot replace matched market-hours historical NBBO/CBBO for every broad and VIX row. The correct use of the broad results is therefore calibrated execution sensitivity: if broad large-cap and VIX spreads behave like the liquid CBBO surface used for imputation, legacy E1 has positive arithmetic results with VIX at \$1 million NAV. The result should not be read as a proof that every historical target contract could have been filled at the imputed quote.

The same distinction applies to margin and assignment. The simulation charges margin drag and assignment penalties, but a simulated penalty is not a broker statement, so execution quality at realized fills is not claimed; establishing it would require market-hours quotes with displayed size, realized fill records, broker margin, and reconciliation against actual statements, all outside the scope of this paper.

Ex-dividend and settlement handling is especially important for options. American single-name options can be assigned early, especially around ex-dividend dates and deep in-the-money short calls. VIX options have their own settlement mechanics and require exact VRO/SOQ handling for expiry P&L. These are not cosmetic accounting details. They can change realized cashflows and margin usage. The simulation therefore prices these events with explicit penalties and settlement labels, and the verification harness requires exact settlement data before VIX rows enter the results.

7. Conclusion

The paper does not establish a new mathematical theory of option portfolios. Premium accounting, Greek Taylor maps, covariance transformation, mean–variance utility, CVaR, and convex constraints are established tools. The repair makes their relationship internally correct: a Greek loading matrix requires factor/residual cross covariance; predictable costs belong in the allocation objective; a utility objective chooses scale; and gross premium is an upper bound that permits cash.

The empirical contribution is correspondingly narrower. R1 is an auditable monthly procedure with point-in-time costs, capacity, margin, collateral, assignment screening, integer execution, and a hard survival gate. The previously reported E1 Sharpes remain useful development history, but E1 was selected after inspection of the post-2020 window and its VIX books have absorbed-zero CPCV wealth paths. Those facts prevent a confirmatory or survival claim regardless of the positive arithmetic Sharpe estimates.

All currently available observations are therefore labeled retrospective development data. A machine-readable freeze manifest records the R1 source hashes, risk policy, data cutoff, and first eligible future decision date. No confirmatory claim is permitted until 36 untouched monthly observations have accumulated. Exact broad and VIX executable quote history, broker margin and assignment records, routed fills, and longer prospective data remain necessary before the procedure can be described as a tradable strategy.

8. R1.1 higher-risk development extension

R1.1 is a separately versioned development arm. It does not replace the frozen R1 artifacts or convert any observation into confirmatory evidence. The policy raises the annualized predicted-volatility ceiling from 15 to 25 percent and uses the smallest positive risk-aversion coefficient consistent with that ceiling. It also records a 50 percent premium-gross target, but does not impose an unconditional gross floor.

The deployment step first solves the ordinary R1 net-utility problem. Only directions selected by that solution and having positive forecast return after their direction-specific cost are eligible for expansion. Holding those signs fixed makes gross linear and prevents offsetting long/short legs in the same contract. The implementation tests whether a 50 percent book satisfies volatility, CVaR, stress, margin, collateral, liquidity, assignment, and concentration limits. If not, it keeps the first-stage book. Thus “target” is not a euphemism for compulsory investment.

Whole-contract execution follows a direct-or-abstain policy. Contract counts in the continuous target are truncated toward zero and the resulting book is rechecked against every hard constraint. If it is feasible, that is the only risky portfolio eligible for execution. If it is infeasible, the strategy holds cash for that period. It does not remove a risk-producing leg, add a VIX hedge, iteratively shrink the book, or substitute a mixed-integer optimum, because those would be different portfolios from the target the continuous model selected. A cash abstention is therefore a valid trading decision rather than an integer-execution failure.

The artifact ledger records both the attempted direct conversion and the selected cash book. Headline exposure and risk diagnostics correctly equal zero during an abstention, while separate fields retain the rejected conversion’s CVaR, stress, volatility, margin, collateral, assignment, liquidity-cap, and base-constraint values. Thus cash reporting no longer erases the reason the target was not traded.

8.1. *VIX-40 close-to-next-open rule*

The risk-off state changes only after an official VIX close. A close at or above 40 sends an exit order at the first executable option quote after the next regular-session open; the first later close below 40 sends a re-entry attempt on the same timing convention. The market observations remain in all later covariance, premia, and EGARCH training windows while the strategy is in cash.

March 1, 2020 was a Sunday. The user reports having decided on that date to exit, but this statement is retrospective and is not generalized to all traders. The VIX had closed at 40.11 on February 28, so the threshold rule and the user-attested instruction both map to Monday, March 2. The event ledger merges them into one order. The March 2 VIX close of 33.42 maps to a March 3 re-entry attempt. Later threshold crossings are handled by the same rule rather than being selectively omitted.

The repository does not presently contain matched March 2/3 and later event-date CBBO for every held equity option. R1.1 therefore generates a licensed-data request manifest and does not score the risk-off return. Long liquidations require executable bids, short covers require executable asks, and displayed size is consumed sequentially. Missing or incomplete fills cannot be replaced by a model price or by the no-rule return. Consequently the risk-off arm is an implemented but currently unscored execution experiment.

8.2. EGARCH diagnostic

The optional diagnostic fits an EGARCH(1,1) model with Student innovations to daily underlying and VX-forward returns using only observations available by each decision date. Its 21-day variance ratio is clipped to $[0.5, 2]$. Delta-factor standard deviations are scaled by the forecast volatility ratio and centered-quadratic factors by the corresponding variance ratio; vega and residual scales remain one. Applying one diagonal transformation to the complete joint factor/residual covariance preserves positive semidefiniteness and the factor/residual cross terms.

Promotion required at least 95 percent forecast coverage, at least a 2 percent pooled QLIKE improvement with a favorable 90 percent block-bootstrap interval, no added survival failure, and no universe-level expected-shortfall deterioration above one NAV percentage point. The checked run achieved 99.9 percent forecast coverage but materially worsened QLIKE and worsened the worst universe's expected shortfall by 1.51 percentage points. EGARCH therefore remains a diagnostic and is not part of the R1.1 candidate.

8.3. Regenerated development evidence

Table 10 reports the chronological January 2018 decision–April 2026 return replay. The longer window is required to expose the March 2020 intervention logic; the separate January 2021–April 2026 comparison remains available in the machine-readable artifacts.

Table 10: R1.1 chronological development replay. Target hit is the fraction of integer books at or above 50 percent premium gross. An infeasible direct conversion produces a cash abstention, not a substitute portfolio or an execution failure. EGARCH is diagnostic only.

Universe	Arm	Obs.	Mean gross	Target hit	Terminal	ES ₉₅	Verdict
larger	EGARCH	93	0.075	0.00	2.736	-0.113	development_survived
larger	R1.1 25%	93	0.067	0.00	2.454	-0.098	development_survived
larger+VIX	EGARCH	93	0.053	0.00	2.758	-0.076	development_survived
larger+VIX	R1.1 25%	93	0.047	0.00	2.804	-0.067	development_survived
orig	EGARCH	93	0.062	0.00	2.418	-0.105	development_survived
orig	R1.1 25%	93	0.055	0.00	2.368	-0.100	development_survived
orig+VIX	EGARCH	93	0.058	0.00	3.018	-0.076	development_survived
orig+VIX	R1.1 25%	93	0.048	0.00	3.160	-0.077	development_survived

Table 11: Whole-contract execution outcomes across the R1.1 base and EGARCH diagnostic books. Direct truncation is selected when feasible; otherwise cash abstention is selected.

Method	Candidates	Feasible	Selected	Mean utility
Cash abstention	744	744	123	0.00000
Direct truncation	744	621	621	0.00376

The 25 percent ceiling increased risk but did not make 50 percent premium gross feasible: the target-hit rate is zero in every universe. Direct conversion is feasible and selected in 621 of the 744 base/EGARCH decisions. The other 123 decisions are explicit cash abstentions: 57 in the two original+VIX arms and 66 in the two larger+VIX arms; neither non-VIX arm

abstains. The attempted book's original breach values remain reported for all 123 decisions. No substitute risky portfolio contributes to returns, and abstention is not counted as an integer-execution or survival failure.

For the non-EGARCH R1.1 arm, annualized geometric return is 11.77 percent in the original universe, 12.28 percent in the larger universe, 16.01 percent with original+VIX, and 14.23 percent with larger+VIX. The corresponding maximum drawdowns are 38.11, 39.40, 17.05, and 12.13 percent. The two non-VIX books never abstain; original+VIX holds cash in 32 of 93 months and larger+VIX in 34 of 93 months. These remain retrospective development results, not confirmatory evidence, and the no-trade convention is part of the newly versioned R1.1 policy rather than a reinterpretation of E1.

These results do not alter the original E1 diagnosis. The legacy VIX-enabled CPCV paths remain absorbed at zero in March 2020 and continue to fail the survival gate. The unscored VIX-40 arm cannot be used to overwrite those paths. A separate R1.1 freeze manifest records the new policy, source hashes, data cutoff, and a fresh requirement of 36 untouched monthly observations before confirmation.

9. Licensed-quote execution audit

The execution audit uses a cost-capped licensed Databento acquisition covering February 2018–April 2026. It completed 1,170 requests and verified 82.0 million rows for \$17.68 against a \$40 cap. The pull contains 10-minute entry-close, next-open, and exit-close windows, full-session VIX intervention days, and decision-date daily volume and open interest.⁴

The quote source is consolidated minute BBO before March 28, 2023 and consolidated MBP-1 thereafter. The purpose is to audit the execution realism of the frozen R1 and R1.1 replays. No return, position, cost input, integerization rule, survival gate, or frozen protocol is changed.

Positions from the frozen weight ledgers are matched to observed NBBO in each leg's window. The audit then replaces only the spread evidence in three scenarios:

- *Mid* retains the frozen non-spread terms: fees, slippage, funding, and the recorded short-call borrow term.
- *Touch* substitutes the observed entry and exit spreads into the frozen cost formula while leaving every other term unchanged.
- *Worst* uses the entry-window maximum spread and the observed exit spread.

Months or legs without observed quotes retain the frozen modeled cost: they are counted, never imputed. Rebuilding modeled cost from the frozen inputs reproduces `predicted_cost` exactly, with a maximum absolute gap below 10^{-16} , so the scenario comparison isolates the spread evidence.

R1.1 weight-aggregate entry coverage is 98.5 percent for larger, 96.2 percent for larger+VIX, 98.7 percent for orig, and 94.1 percent for orig+VIX. Round-trip coverage is 62.9, 29.1, 65.1, and 25.5 percent, respectively. VIX options are cash settled at the SOQ, so settlement-held positions have no exit-close quote. Where the cached holding path contains a last-tradable-session quote, the audit uses it as a fallback and labels that source in the ledger.

Table 12: R1.1 retrospective development returns under frozen modeled costs and three licensed-quote execution scenarios. Coverage is aggregated by absolute portfolio weight.

Config	modeled ann.	modeled Sortino	modeled maxDD	mid ann.	mid Sortino	mid maxDD	touch ann.	touch Sortino	touch maxDD	worst ann.	worst Sortino	worst maxDD	Entry cov. %	Round-trip cov. %
larger	0.123	1.114	-0.394	0.130	1.176	-0.391	0.066	0.664	-0.413	0.057	0.590	-0.418	98.472	62.902
larger+VIX	0.142	2.197	-0.121	0.145	2.254	-0.121	0.121	1.812	-0.128	0.119	1.771	-0.129	96.157	29.134
orig	0.118	1.097	-0.381	0.123	1.145	-0.378	0.086	0.830	-0.389	0.080	0.784	-0.390	98.696	65.061
orig+VIX	0.160	2.275	-0.171	0.163	2.322	-0.167	0.146	2.052	-0.185	0.145	2.029	-0.186	94.096	25.504

Mid fills are marginally better than the modeled series. At the touch, however, the no-VIX orig book declines from 0.118 to 0.086 annualized return and from 1.097 to 0.830 Sortino. The orig+VIX book moves from 0.160 to 0.146 annualized return and from 2.275 to 2.052 Sortino; its worst scenario remains at 0.145 annualized return and 2.029 Sortino. Observed spreads for the broad equity books are wider than the inferred-CBBO proxy assumed in the frozen costs. Even so, the headline orig+VIX configuration survives the harshest observed-quote scenario with Sortino approximately 2.0.

⁴The operational acquisition record, including the private-request-ledger hash, is `analysis/artifacts/databento_audit/databento_acquisition_summary.json`.

Table 13: Modeled and observed relative-spread quantiles before and after the quote-schema cutover.

Arm	Regime	Source	p25	p50	p75	p90
R1	post_2023_03_28	modeled	0.012	0.015	0.017	0.030
R1	post_2023_03_28	observed	0.028	0.046	0.078	0.133
R1	pre_2023_03_28	modeled	0.011	0.016	0.017	0.034
R1	pre_2023_03_28	observed	0.013	0.026	0.049	0.081
R1.1	post_2023_03_28	modeled	0.011	0.015	0.016	0.024
R1.1	post_2023_03_28	observed	0.026	0.044	0.073	0.124
R1.1	pre_2023_03_28	modeled	0.013	0.017	0.018	0.027
R1.1	pre_2023_03_28	observed	0.014	0.024	0.042	0.071

For R1.1 before March 28, 2023, the median observed relative spread is 2.4 percent versus 1.7 percent modeled, and the observed 90th percentile is 7.1 percent versus 2.7 percent modeled. After the cutover, the corresponding observed/model medians are 4.4/1.5 percent and the 90th percentiles are 12.4/2.4 percent. The split distinguishes minute-observable consolidated BBO from the later consolidated MBP-1 record rather than treating the two schemas as identical evidence.

Decision-mark accuracy is stable across the cutover. For R1.1, median absolute relative error against observed entry mids is 2.9 percent before March 28, 2023 and 3.0 percent after; the corresponding 90th percentiles are 13.8 and 13.0 percent.

Table 14: Entry-date contract participation and capacity diagnostics. Volume is reported against summed and maximum publisher observations; open interest is a separate diagnostic.

Arm	volume_sum_p25	volume_sum_p50	volume_sum_p75	volume_sum_p90	volume_max_p25	volume_max_p50	volume_max_p75	volume_max_p90	open_interest_p25	open_interest_p50	open_interest_p75	open_interest_p90	optimizer_005_breaches	volume_010_breaches	oi_002_breaches
R1	0.001	0.002	0.008	0.020	0.006	0.020	0.048	0.000	0.000	0.001	0.004	0.004	83	27	54
R1.1	0.001	0.004	0.012	0.028	0.004	0.011	0.023	0.069	0.000	0.001	0.002	0.005	186	57	140

R1.1's 90th-percentile participation is 2.8 percent of summed decision-date volume, 6.9 percent against the maximum publisher observation, and 0.5 percent of open interest. The 5 percent optimizer-volume, 10 percent capacity-volume, and 2 percent open-interest diagnostics record 83/27/54 R1 breaches and 186/57/140 R1.1 breaches. These checks cover entry dates only, not the holding path. R1 contract counts are implied at \$1 million NAV through the R1.1 symbol lookup and are identified by the `contracts_source` column; R1.1 counts are the frozen integerized contracts.

The intervention artifact contains 156 evidence-only position rows, of which 154 have a valid quote aggregate. Median quote-presence is 99.7 percent; across rows with quotes, the median of the session median spreads is 2.7 percent and its cross-row 90th percentile is 12.9 percent. The cross-contract medians of first-valid and full-session VWAP touch are 5.250 and 4.531. These are post-freeze feasibility observations only. The risk-off arm remains unscored under the frozen protocol.

The remaining limitations are:

- pre-2023 option quotes are minute-observable, not tick-exact;
- no assignment/early-exercise validation;
- no corporate-action adjustments;

- volume/OI validated at entry dates only;
- quotes prove executability at displayed top-of-book size, not realized fills;
- the risk-off intervention arm remains unscored.

The audit strengthens cost realism of the retrospective development evidence; it does not convert it into confirmatory evidence, and the 36-month prospective freeze stands.

10. R2 robust-Sortino development strategy

R2 is a separately versioned retrospective development strategy. It does not alter the frozen R1 or R1.1 sources, artifacts, or freeze manifests, and it replaces R1.1 only if every prespecified historical and simulation gate passes. Its Sortino target is zero net return: because cash earns zero in this comparison, every negative post-cost month enters downside deviation.

At each decision, R2 estimates two complete Greek factor/residual joint covariances. The recent estimate uses the last 36 observed monthly returns; the expanding estimate uses all prior observations from February 2015. Candidate recent weights of 0.25, 0.50, and 0.75 are selected by inner chronological covariance QLIKE. A one-standard-error rule selects the lowest recent weight within the best estimate's uncertainty band, and the weight is fixed at 0.50 until 12 inner forecasts exist. Both estimates retain the factor/residual cross block; their joint covariance matrices are blended before transformation to option space.

Missing option returns are not set to zero. A contract requires at least 24 observations in the recent 36-date window. Missing cells receive the contemporaneous Greek-factor prediction, and five deterministic scenario panels add residual draws resampled from that contract's observed residuals. R2 evaluates the worst downside deviation across the recent panel, an expanding six-month circular-block panel, and all five imputation panels. The structural R1 premia channels remain in place, but the underlying and volatility premia use expanding exponential weights with a 36-month half-life; the direct historical option-mean weight remains zero.

Daily data condition risk, not option P&L. For each surviving underlying, training data are limited to the last 756 daily observations, with at least 500 required. HAR realized variance and EWMA forecasts are combined over weights from zero to one in quarter steps by rolling QLIKE, again with a one-standard-error preference for the more stable EWMA model. The 21-day variance ratio is clipped to $[0.67, 1.50]$. One diagonal transform scales delta factor standard deviations by the volatility ratio and centered-quadratic factors by the variance ratio; vega and residual dimensions are unchanged. This single joint transform preserves positive semidefiniteness and the factor/residual cross terms. Missing daily history produces an audited unscaled-covariance fallback.

The first optimization stage is a robust SOCP. It maximizes expected return net of direction-specific predictable costs subject to unit worst-family zero-target downside deviation. It uses one signed weight vector with abs/positive-part expressions; there are no overlapping positive and negative legs that can manufacture gross. Because the ratio determines direction but not size, a separate scalar stage normalizes the direction and maximizes expected net log growth. Scale zero is admissible and is selected when no positive scale has positive expected log growth.

The scalar stage applies a 10 percent annualized downside-deviation ceiling and the R1.1 25 percent annualized volatility ceiling. It also retains the 10 percent monthly CVaR, 20 percent named stress loss, 75 percent short-margin, 100 percent collateral, assignment, liquidity, Greek, beta, and concentration limits. The worst training three-month and six-month compounded losses may not exceed 15 and 20 percent, respectively. The continuous target is converted directly to whole contracts by truncating counts toward cash. If that exact book breaches any constraint, the selected position is cash; no substitute portfolio is introduced, and the rejected book's original diagnostics remain in the abstention ledger.

Table 15: R2 chronological retrospective development replay, February 2018–April 2026. Sharpe is secondary; the promotion decision also uses path survival and the locked simulation suites described in the text.

Universe	N	Geom. return	Sortino	Max DD	Abstain	Verdict
larger	93	0.006	0.479	-0.043	0	development_survived
larger+VIX	93	-0.012	-0.265	-0.127	0	development_survived
orig	93	0.074	1.983	-0.120	1	development_survived
orig+VIX	93	0.023	0.397	-0.170	1	development_survived

The VIX-40 overlay remains excluded from headline R2 returns until complete executable OPRA event quotes exist. It therefore cannot overwrite either R2's realized path or the original E1 absorbed-zero finding. R2's 5,000 six-month circular-block paths, 2,000 60-month repriced paths under both joint-GARCH blocks and Gaussian-copula sensitivity, and 200 re-fitted histories per universe are locked diagnostics. CPCV remains a dependence and fragility diagnostic rather than a tradable or confirmatory claim. A separate prospective freeze requires 36 untouched future monthly observations before any confirmatory statement.

A. Technical appendix

The main text keeps the proof sketches close to the economic argument. This appendix records standard mathematical details used by the implementation. They are included for auditability, not as claims of new theory. The maximum-Sharpe discussion describes the legacy E1 geometry and why R1 replaces it with strictly concave net utility.

A.1. Scale, costs, and no-free-exposure logic

Proposition A.1 (No free option exposure). *Let $\mathcal{C}$ include binding gross-premium, short-premium, stress-loss, margin/collateral, or contract-level liquidity controls, and assume the imposed controls jointly separate exposure: any premium-weight direction on which every imposed control vanishes carries no premium, payoff, or Greek exposure. Then no economically nonzero sequence of option positions can take unbounded premium, payoff, or Greek exposure while remaining feasible and using zero capital.*

Proof. The listed controls are positively homogeneous in the premium-weight vector. If $\|w_k\| \rightarrow \infty$ along an economically nonzero direction, at least one controlled premium, margin, stress, Greek, or liquidity functional diverges, violating feasibility at some finite scale. The only directions along which no imposed control diverges lie in the joint null space of the controls, and by the separation assumption those directions carry no premium, payoff, or Greek exposure, so they are not economically nonzero. Therefore every economically meaningful option direction must consume a real budget before scale can diverge. $\square$

This proposition is one reason the paper does not treat gross option Sharpe as a trading claim. A cheap option can carry large return exposure per dollar of premium, and a short option can receive premium while creating a future liability. The optimizer is only an allocation rule after premium, short-premium, stress, margin, concentration, and liquidity limits turn the scale-free return object into a feasible book.

Proposition A.2 (Operational monotonicity). *For*

$$V(\mathcal{C}, c) = \max_{w \in \mathcal{C}} \hat{\mu}^\top w - \frac{\gamma}{2} w^\top \Sigma w - c^\top |w|,$$

with $\Sigma \succeq 0$, if $\mathcal{C}_1 \subseteq \mathcal{C}_0$ and $c_1 \geq c_0$ componentwise, then $V(\mathcal{C}_1, c_1) \leq V(\mathcal{C}_0, c_0)$.

Proof. For every $w \in \mathcal{C}_1$, the same w is feasible for $\mathcal{C}_0$, and $-c_1^\top |w| \leq -c_0^\top |w|$. Maximizing a weakly lower objective over a weakly smaller feasible set cannot increase the value. $\square$

The monotonicity result justifies the conservative reporting convention. Higher spread, fee, slippage, margin, assignment, or impact assumptions can only make the utility value worse. Tighter displayed-size, stress, or margin budgets can only make the feasible set smaller. Positive results under full costs and hard caps are therefore more meaningful than positive gross results, while failures under exact execution data would supersede the current proxy-spread sensitivity.

A.2. Mean estimation error, shrinkage, and purge details

Proof of Proposition 2.4. Each error e_i is sub-Gaussian with variance proxy $\sigma_i^2/T \leq \bar{\sigma}^2/T$, so for every $\lambda > 0$, $\mathbb{E} \exp(\lambda e_i) \leq \exp(\lambda^2 \bar{\sigma}^2/2T)$ and the same holds for $-e_i$. Let $M =$

$\max_i |e_i|$, the maximum of the $m = 2n$ variables $\{\pm e_i\}$. By Jensen's inequality and a union bound on the moment generating function,

$$\exp(\lambda \mathbb{E}[M]) \leq \mathbb{E} \exp(\lambda M) \leq \sum_{j=1}^m \mathbb{E} \exp(\lambda X_j) \leq m \exp\left(\frac{\lambda^2 \bar{\sigma}^2}{2T}\right),$$

so $\mathbb{E}[M] \leq \log(m)/\lambda + \lambda \bar{\sigma}^2/2T$. Optimizing at $\lambda = \sqrt{2T \log m}/\bar{\sigma}$ gives $\mathbb{E}[M] \leq \bar{\sigma} \sqrt{2 \log(2n)/T}$, the first display. For the second, Hölder's inequality gives $\sup_{w \in \mathcal{C}} e^\top w \leq \sup_{\|w\|_1 \leq g} e^\top w = g \|e\|_\infty = gM$; take expectations. For the objective-loss statement, write the optimizer's concave criterion as $F_\mu(w) = \mu^\top w - R(w)$, where R collects the risk and cost penalties, and let $\tilde{w}$ maximize $F_{\hat{\mu}}$ and w^* maximize F_μ over $\mathcal{C}$. Then

$$F_\mu(\tilde{w}) \geq F_{\hat{\mu}}(\tilde{w}) - g \|e\|_\infty \geq F_{\hat{\mu}}(w^*) - g \|e\|_\infty \geq F_\mu(w^*) - 2g \|e\|_\infty,$$

so the expected sacrifice of true objective value is at most $2g \mathbb{E} \|e\|_\infty \leq 2g \bar{\sigma} \sqrt{2 \log(2n)/T}$. $\square$

Proof of Corollary 2.5. The sandwich argument above holds verbatim when the mean input is the structural forecast: with deterministic bias vector $b = \mu^{\text{struct}} - \mu$ and $\|b\|_\infty \leq \beta$, the loss bound is $2g\beta$. Comparing the two worst-case bounds, the historical mean has the smaller guarantee exactly when $\bar{\sigma} \sqrt{2 \log(2n)/T} < \beta$, which rearranges to $2 \log(2n) < T \beta^2 / \bar{\sigma}^2$, i.e. $n < \frac{1}{2} \exp(T \beta^2 / 2 \bar{\sigma}^2)$. At fixed T the left side grows in n while the right side is constant, which yields the stated crossover. $\square$

The bounds compare worst-case guarantees, not realized losses, and the sub-Gaussian model ignores the correlation and missingness of real option panels; both simplifications favor the historical mean, so the crossover argument is conservative in the direction the paper uses it.

Lemma A.3 (One-month purge suffices for monthly option labels). *Index each label by its return month-end. Assume every tested contract's information window spans at most Δ calendar days and ends at its return month-end, where $\Delta \approx 44$ days is the maximum decision-to-expiry span of the monthly-tenor contracts. If the realized calendar distance $|t_e - t_r|$ between every training return month-end t_r and every test return month-end t_e is at least G with $G > \Delta$ in every fold—a two-sided requirement, because combinatorial folds train on months both before and after each tested block—then no training label's information window overlaps any test label's window.*

Proof. A training label with return month-end t_r has window contained in $[t_r - \Delta, t_r]$; a test label with return month-end t_e has window contained in $[t_e - \Delta, t_e]$. If $t_r \leq t_e$, overlap requires $t_r \geq t_e - \Delta$; if $t_r > t_e$, overlap requires $t_r - \Delta \leq t_e$. Either way overlap requires $|t_e - t_r| \leq \Delta$. The purge/embargo schedule enforces $|t_e - t_r| \geq G > \Delta$ for every training/test pair, so the windows are disjoint. $\square$

The quantity G is not assumed; it is measured. The verifier recomputes the minimum gap across all folds, schemes, and configurations from the fold schedule and ledger artifacts—on

both sides of every test block, since the pre-test boundary is set by the purge alone while the post-test boundary is set by the purge plus embargo—and asserts $G > 44$ days; in the current artifacts the realized minimum is $G = 59$ days, attained on the pre-test side. The one-month purge plus one-month embargo is therefore sufficient with two weeks of slack, and the sufficiency claim is re-audited on every verification run rather than hand-typed.

A.3. Legacy maximum Sharpe and conic reduction

Let $\mu = \hat{\mu}_t$ and $\Sigma = \hat{\Sigma}_{O,t}$. When $\Sigma \succ 0$, the unconstrained maximum-Sharpe direction is the usual tangency direction even though the assets are option cashflows:

$$\arg \max_{w \neq 0} \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \propto \Sigma^{-1} \mu.$$

The difference from stock Markowitz is not the algebra of the tangency direction; it is the construction of μ , Σ , and the feasible set. Option premium weights have cash and collateral as the numeraire, so the unit-stock-budget equation is replaced by gross premium, short-premium, stress, margin, and liquidity budgets.

Proposition A.4 (Conic maximum-Sharpe form). *Let $K \subseteq \mathbb{R}^n$ be a closed convex cone containing some w with $w^\top \mu > 0$, and let $\Sigma \succ 0$. Then*

$$\sup_{w \in K \setminus \{0\}} \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} = \max\{y^\top \mu : y \in K, y^\top \Sigma y \leq 1\}.$$

The right side is a convex quadratically constrained program and is a second-order-cone program when a Cholesky or square-root factor of Σ is used.

Proof. For any nonzero $w \in K$, define $y = w/\sqrt{w^\top \Sigma w}$. Since K is a cone, $y \in K$, $y^\top \Sigma y = 1$, and $y^\top \mu$ equals the Sharpe ratio of w . This shows that the right side is at least the left. Conversely, any feasible y with $y^\top \mu > 0$ satisfies

$$\frac{y^\top \mu}{\sqrt{y^\top \Sigma y}} \geq y^\top \mu,$$

so the left side is at least the right. Positive definiteness makes the risk ball bounded, and the objective is linear on a closed convex feasible set. $\square$

This geometry determines only a direction. Legacy E1 imposed gross equality on positive and negative split variables, which allowed overlapping legs to satisfy the algebra while the net economic book remained smaller. R1 does not use that representation. It solves strictly concave net utility over one position vector, imposes $\|w\|_1 \leq g$, and allows the unallocated balance to remain cash.

A.4. Joint risk-model estimator

Let $Z_t = [f_t^\top, \varepsilon_t^\top]^\top$, where $\varepsilon_t = r_t - B^G f_t$. R1 standardizes the aligned training columns, applies Ledoit–Wolf shrinkage to their joint correlation-scale covariance, rescales to the original

units, and partitions the result as

$$\hat{S}_Z = \begin{bmatrix} \hat{\Omega} & \hat{\Gamma} \\ \hat{\Gamma}^\top & \hat{\Sigma}_\varepsilon \end{bmatrix}.$$

The option covariance is

$$\hat{\Sigma}_O = [B^G \ I] \hat{S}_Z [B^G \ I]^\top.$$

This transformation is PSD whenever $\hat{S}_Z$ is PSD and retains the cross terms. With shrinkage disabled it reproduces the aligned sample covariance of option returns. Only after this joint transformation does the implementation apply a small numerical eigenvalue floor. Legacy E1's diagonal residual covariance remains in the archived development path, not in R1.

A.5. Appendix use and audit trail

Mathematical mechanics that strengthen the results but would slow the main reading path are kept in this appendix. Large empirical ledgers remain machine-readable artifacts under `analysis/artifacts/breadth_solutions/robustness/`; the body prints only the scoreboard, distributional summaries, robustness heatmap, capacity/spread panel, and rolling OOS paths. Every number in the body traces to one of those ledgers, and the verification harness recomputes the headline statistics from the artifacts on each build.

Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. Raw licensed OPRA/Databento data are not redistributed. The repository instead ships the audit's aggregate artifacts under `analysis/artifacts/execution_audit/` and the acquisition summary with request hashes. Entry-window costs and decision marks for held positions are audited against licensed consolidated quotes. The baseline eight equity-option spread rows use local historical panel CBBO. Broad-name and VIX spread rows that lack matched historical CBBO use the documented inferred CBBO proxy and are labeled as execution-sensitivity evidence. Exact VRO/SOQ settlement files are versioned by source hash before VIX option rows enter the reported tables.

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